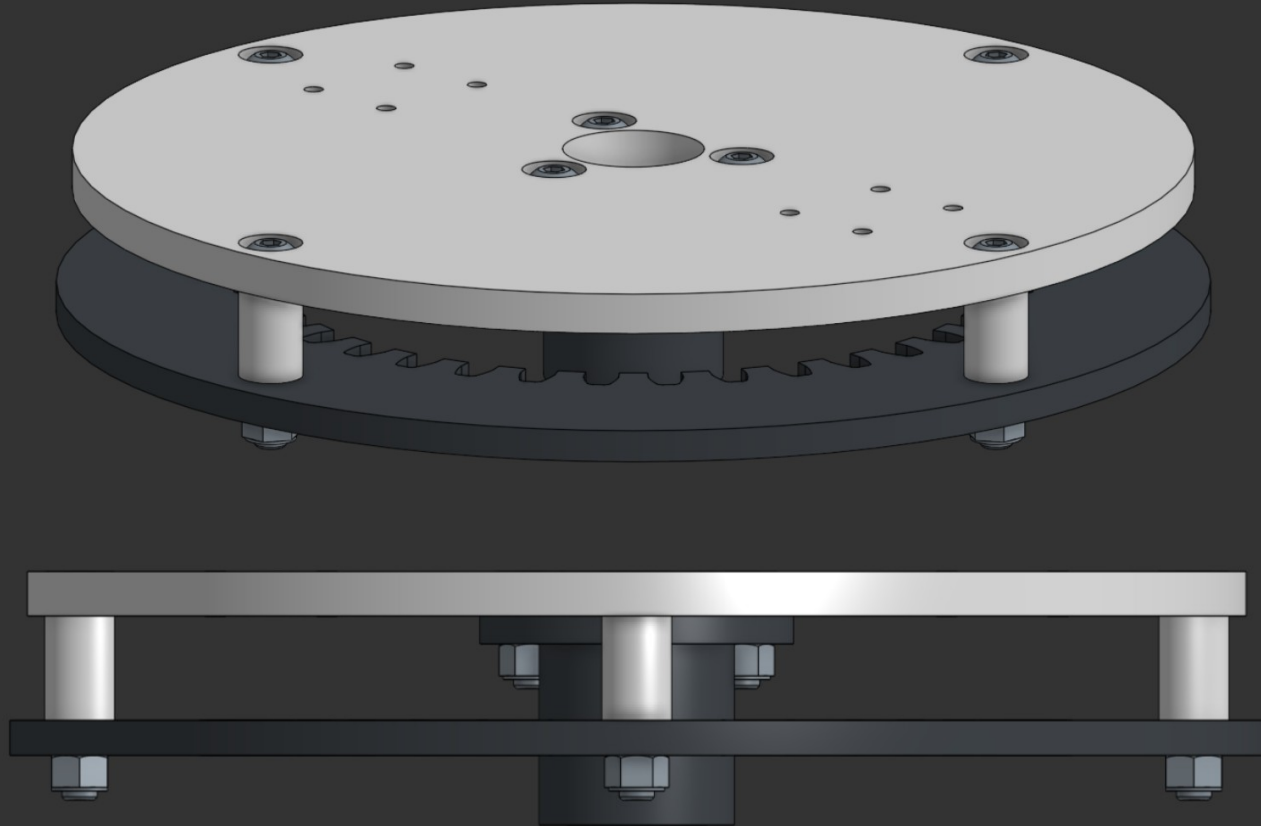


Arm Base Plate Assembly Instructions



Parts

Major Components:

Arm Base Plate

Wire Holder

36T Ring Gear

4x M5x10x20mm Spacer

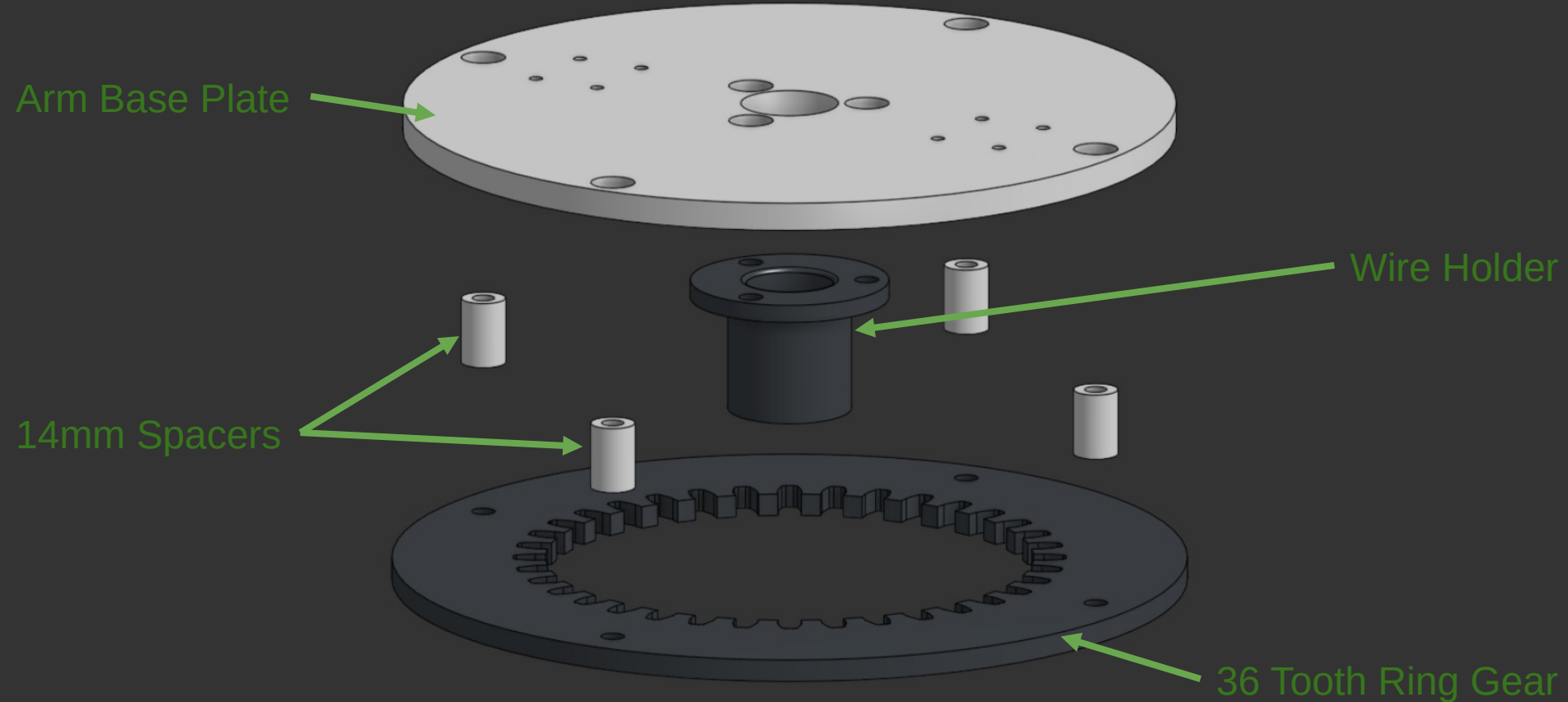
Minor Components:

4x M5x30mm Button Head Bolt

3x M5x14mm Button Head Bolt

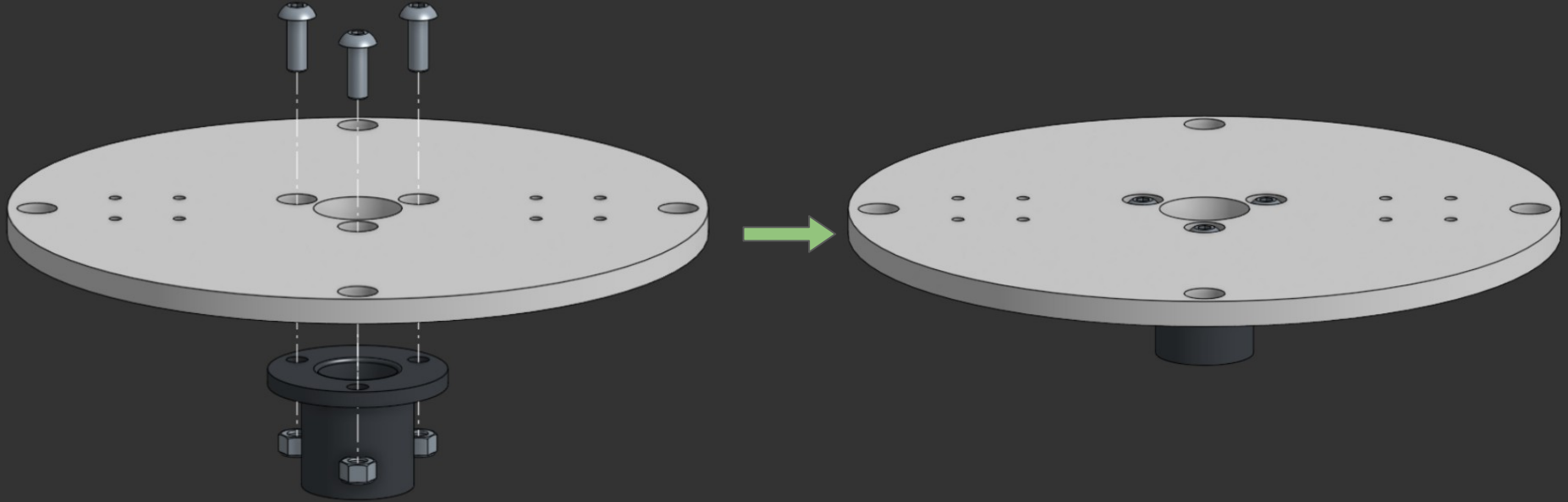
7x M5 Nylock Nuts

Major Components



Step 1

Bolt the Wire Holder through the center of the Arm Base Plate



Parts

-Arm Base Plate

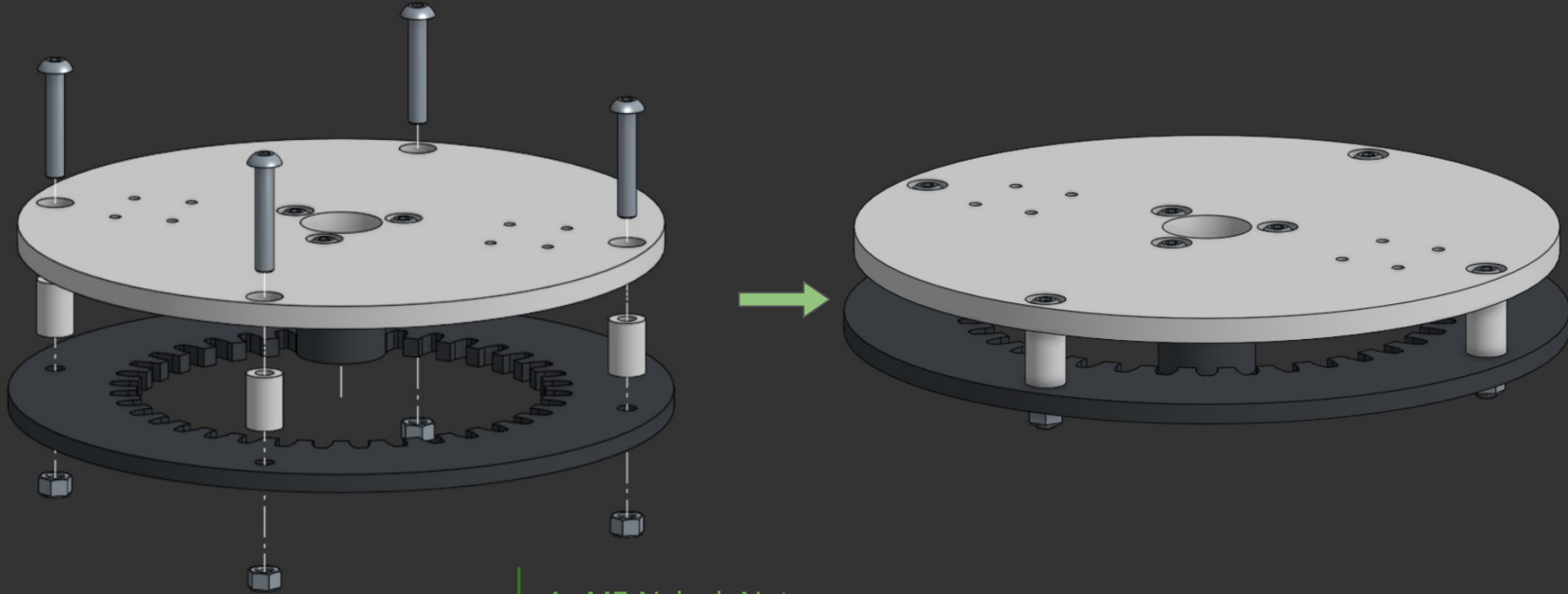
-Wire Holder

-3x M5x14mm Button Head Bolts

-3x M5 Nylock Nuts

Step 2

Bolt the Arm Base Plate to the 36T Ring Gear with 20mm spacers in between



Parts

-Previous step

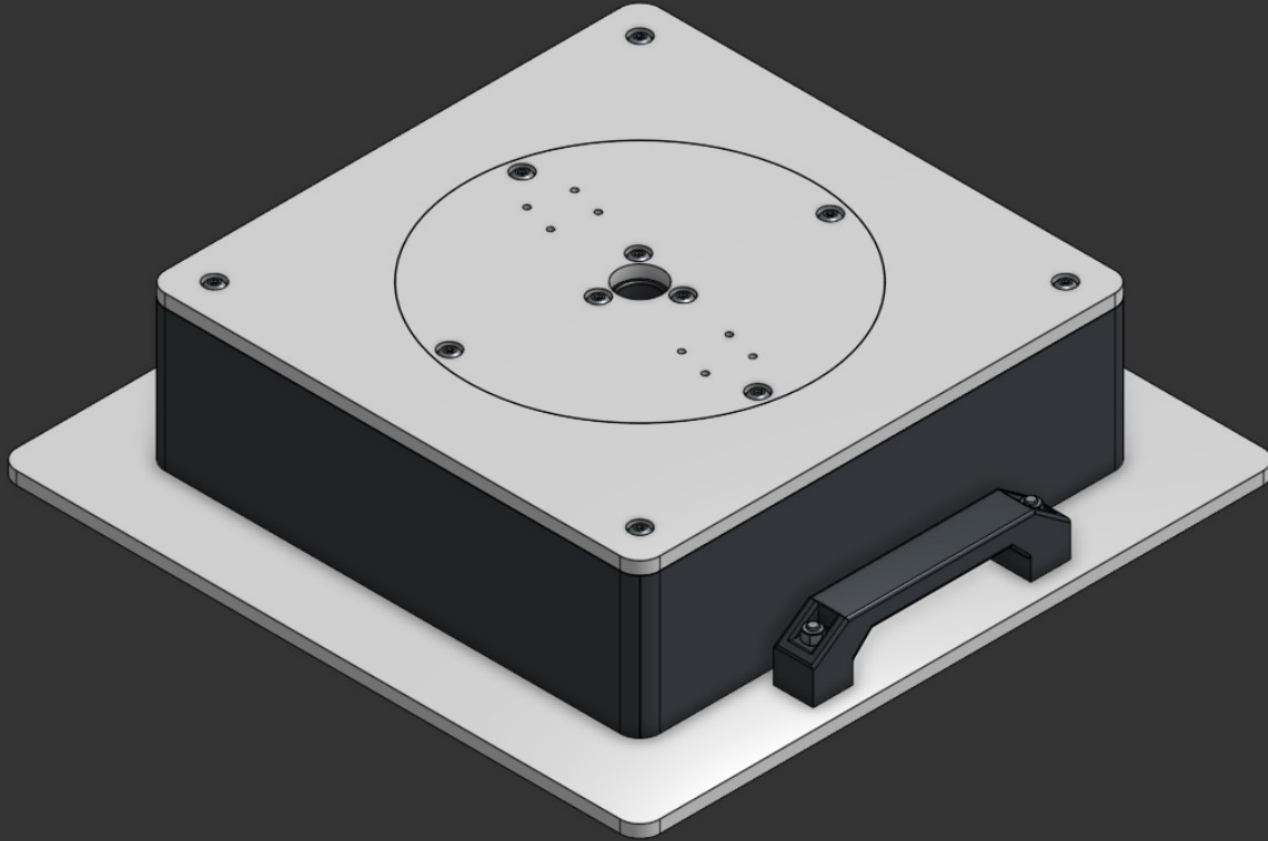
-4x M5x10x20mm Spacer

-4x M5x30mm Button Head Bolts

-4x M5 Nylock Nuts

-36T Ring Gear

Arm Base Assembly Instructions



Parts

Major Components:

Arm Base Plate Assembly

Base Plate

Structure Plate

Base Top Plate

4x Base Wall

2x Base Handle

12T Ring Gear

Robotis XM540-W270-T

Actuator

Parts Continued

Minor Components:

20x Nylock Nut

8x M5x14mm Button Head Bolt

4x M5x30mm Button Head Bolt

8x M5x30mm Socket Head Bolt

8x M5x60mm Button Head Bolt

4x M2.5x12mm Socket Head Bolt

4x M2.5x25mm Socket Head Bolt

8x 5x11x5mm Bearing

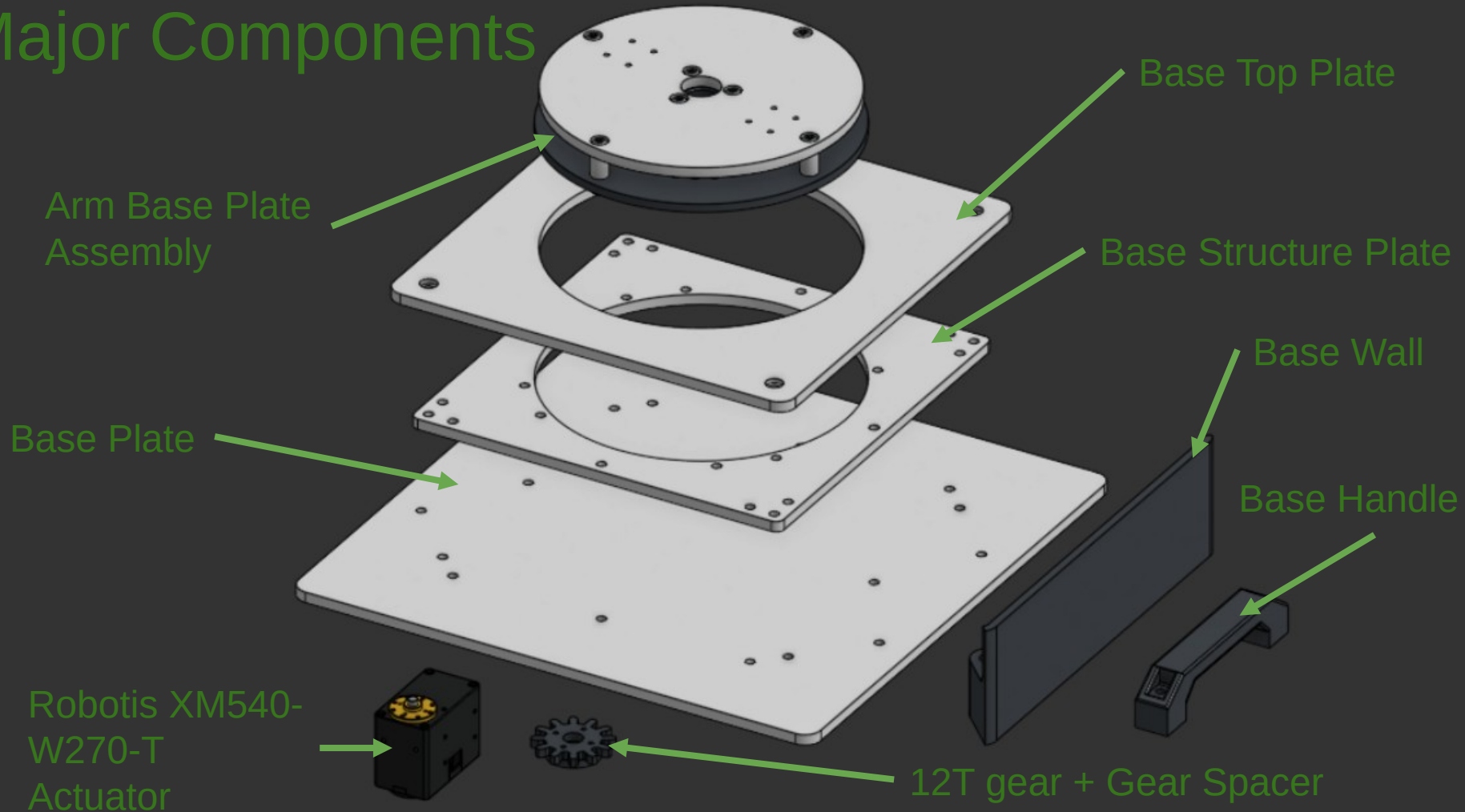
16x 5x16x5mm Bearing

4x 5x8x25mm M5 Threaded
Spacers

8x 5x10x40mm M5 Threaded
Spacers

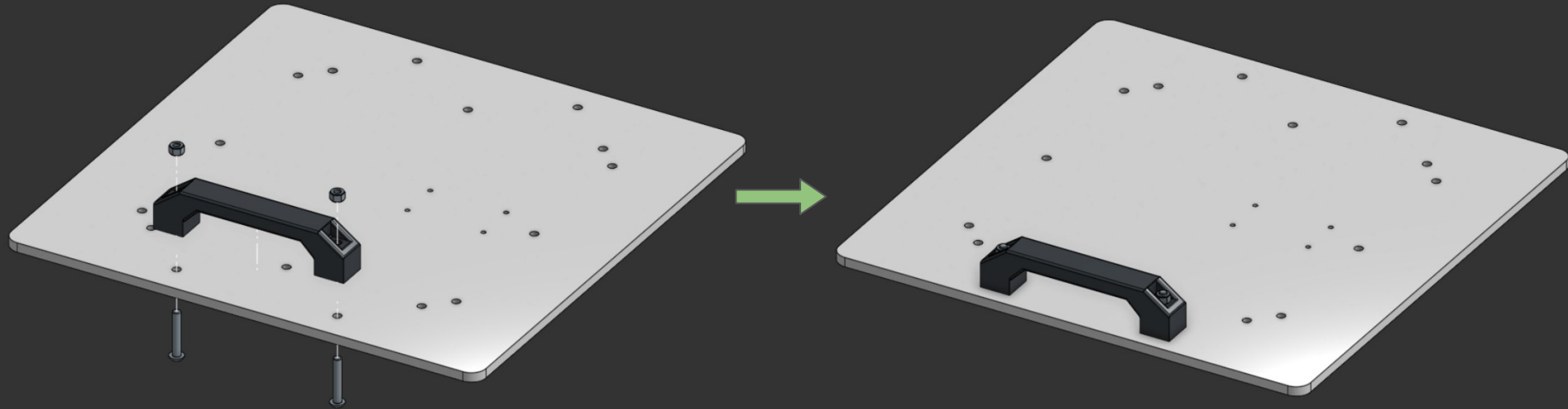
4x Robotis Spacers

Major Components



Step 1

Bolt the Base Handle to the edge of the Base Plate



Parts

-Base Plate

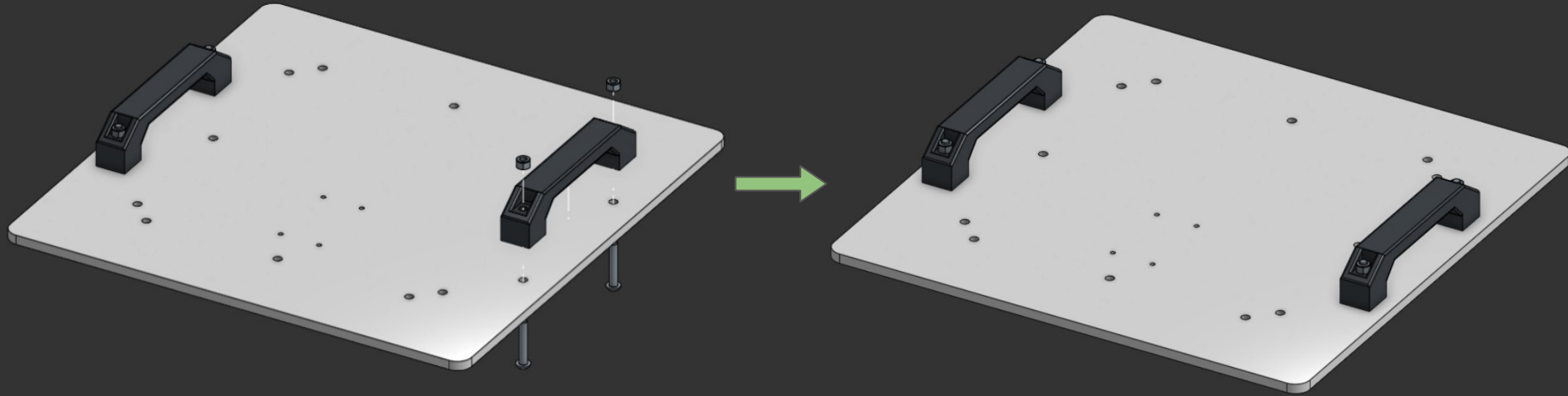
-Base Handle

-2x M5x30mm Button Head Bolt

-2x Nylock Nuts

Step 2

Repeat for the other side



Parts

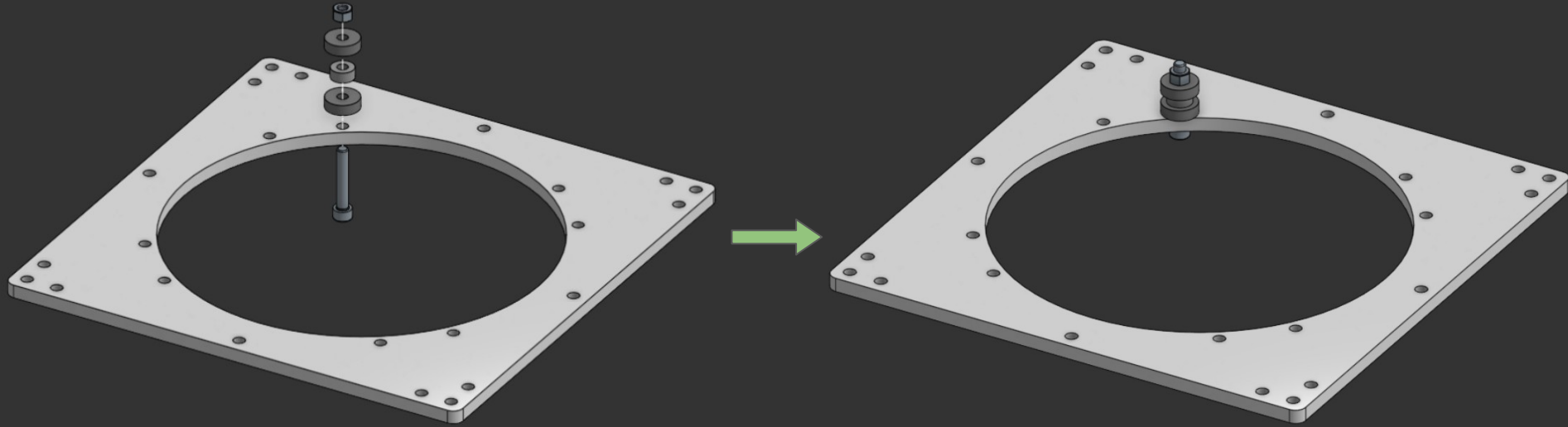
-Previous step

-Base Handle

-2x M5x30mm Button Head Bolt

-2x Nylock Nuts

Step 3 Bolt a bearing stack (in the order of wide, thin, wide) around the center of the structure plate

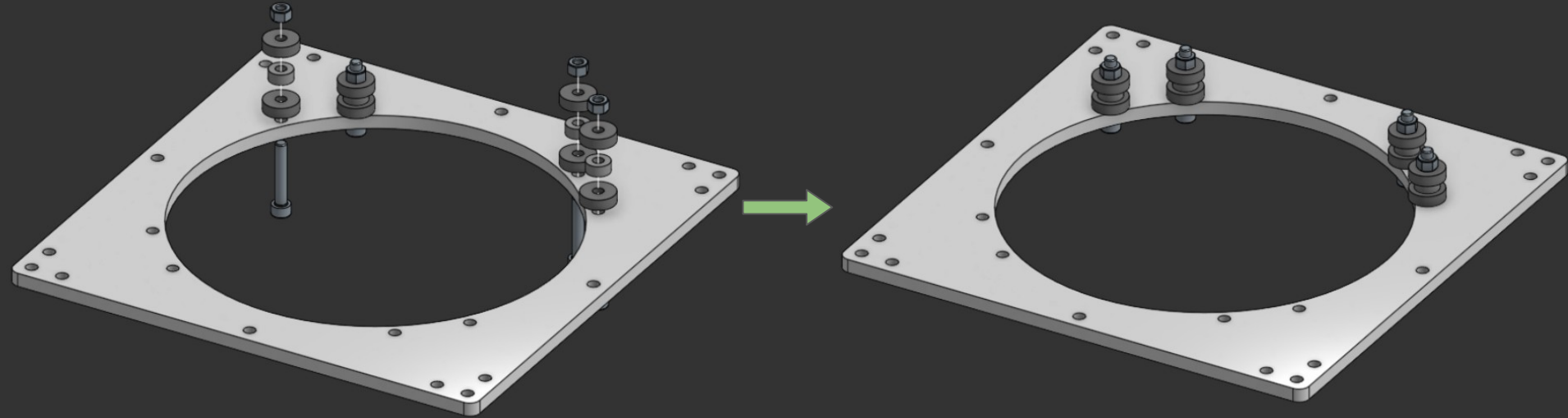


Parts

- Structure Plate
- 2x 5x16x5mm Bearing
- 5x11x5mm Bearing

- M5x30mm Hex Socket Head Bolt
- Nylock Nut

Step 4 Bolt 3 more bearing stacks around one half of the center of the structure plate

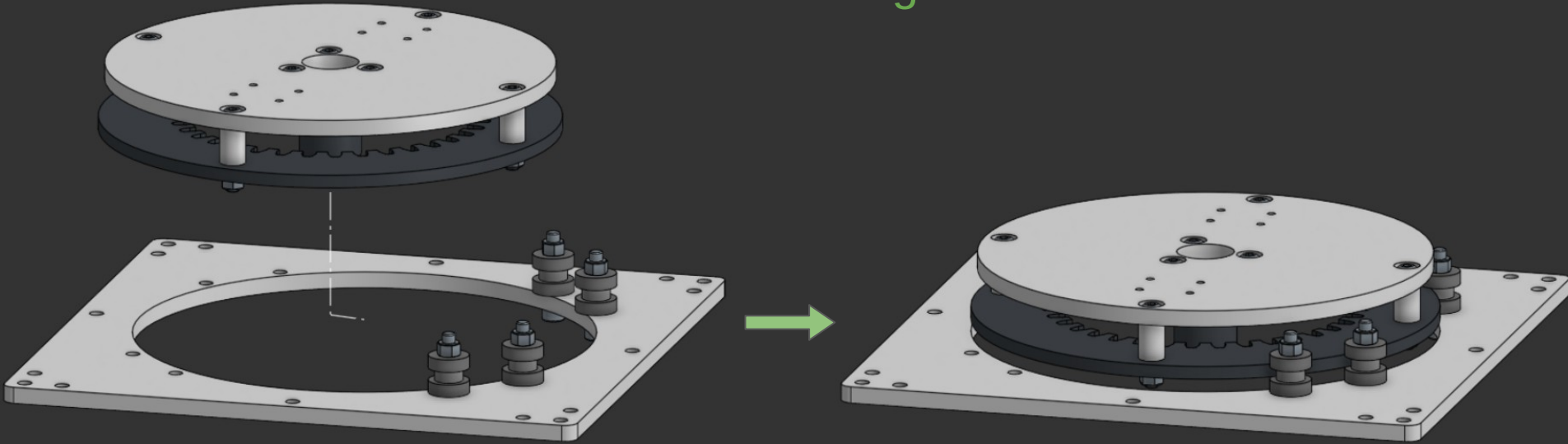


Parts
-Previous step
-6x 5x16x5mm Bearing
-3x 5x11x5mm Bearing

-3x M5x30mm Hex Socket Head Bolt
-3x Nylock Nut

Step 5

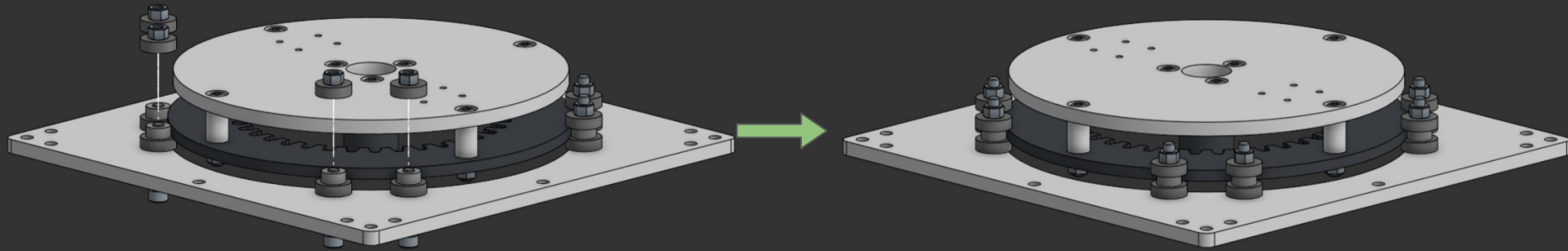
Insert the Arm Base Assembly between the bearings in the bearing stacks



- Parts
- Previous step
 - Arm Base Assembly

Step 6

Bolt four more bearing stacks around the other half of the structure plate



Parts

-Previous step

-8x 5x16x5mm Bearing

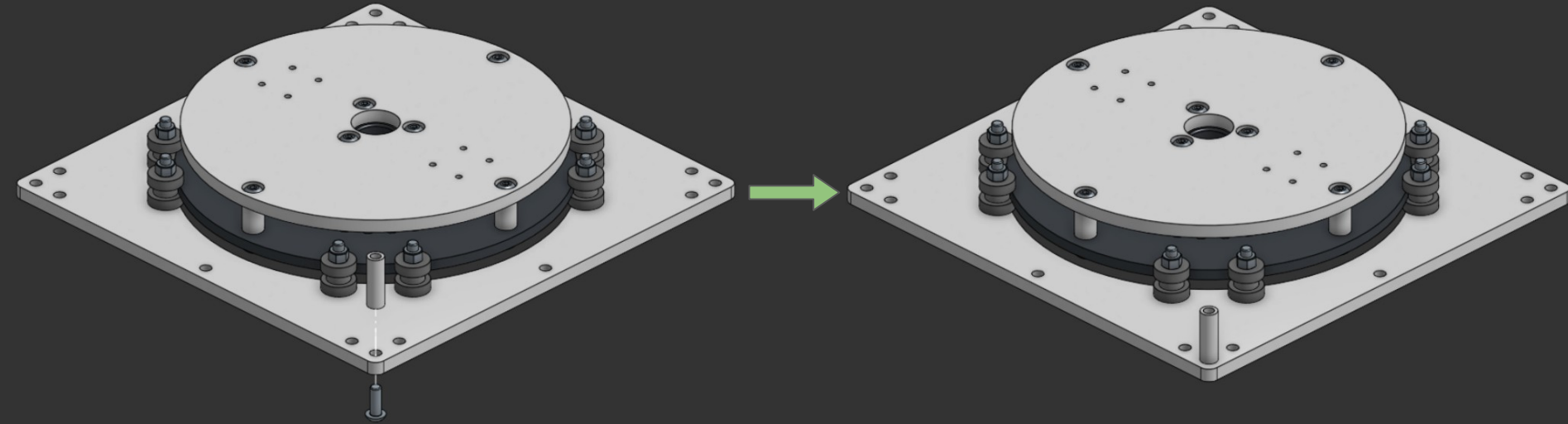
-4x 5x11x5mm Bearing

-4x M5x30mm Hex Socket Head Bolt

-4x Nylock Nut

Step 7

Bolt a threaded spacer onto one corner of the Structure Plate

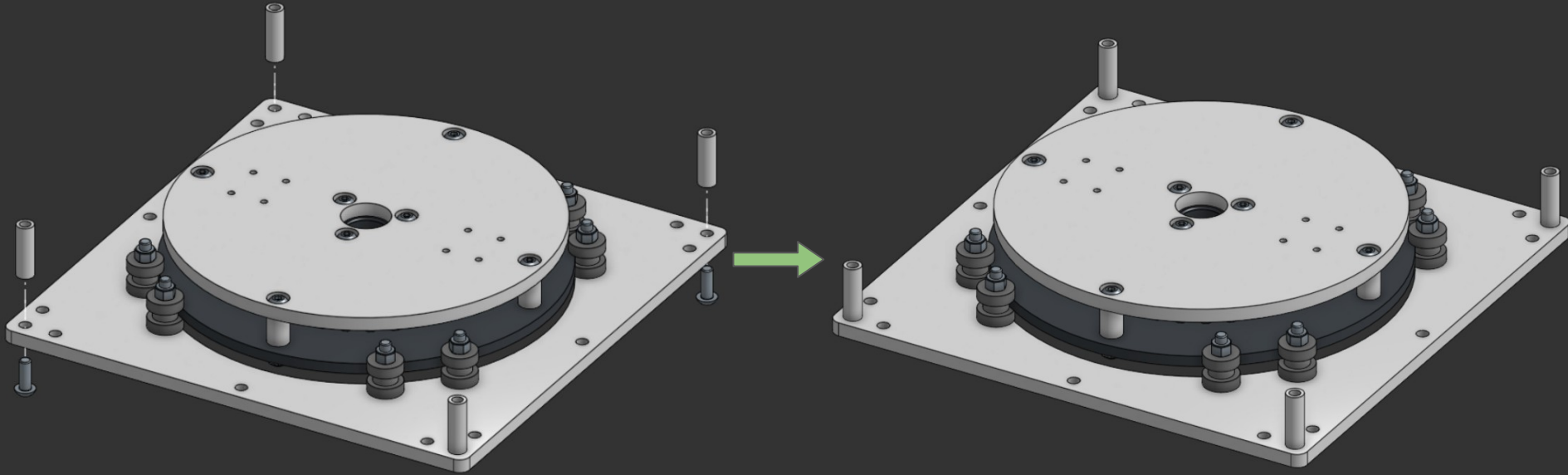


Parts

- Previous step
- 5x8x25mm M5 Threaded Spacers
- M5x14mm Button Head Bolt

Step 8

Bolt threaded spacers on all other corners of the Structure Plate

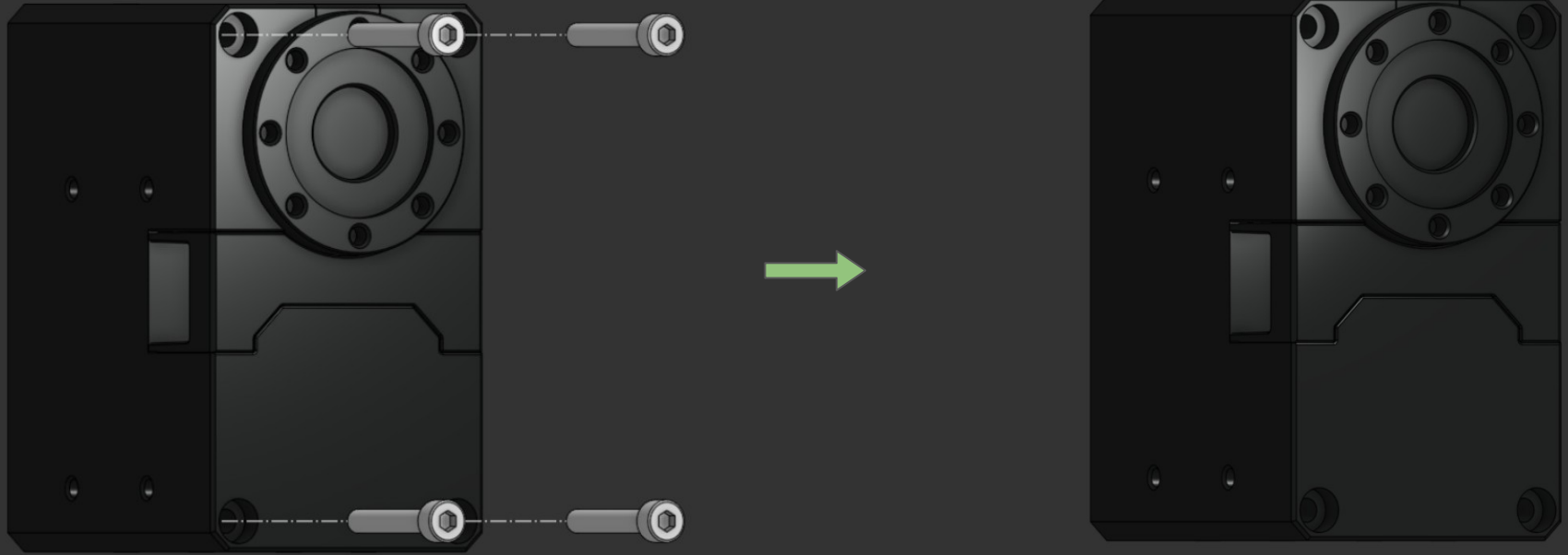


Parts

- Previous step
- 3x 5x8x25mm M5 Threaded Spacers
- 3x M5x14mm Button Head Bolt

Step 9

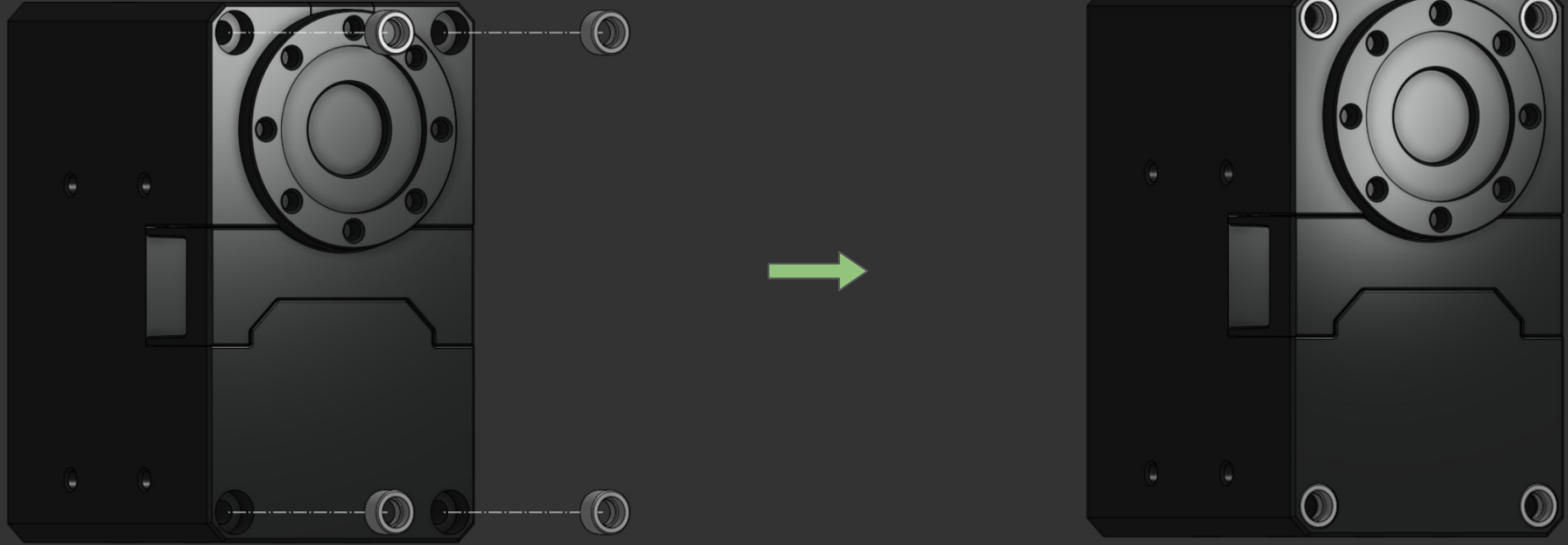
Remove the four bolts on the back of the actuator



Parts

-Robotis XM540-W270-T Actuator

Step 10 Insert four Robotis Spacers into the back of the actuator

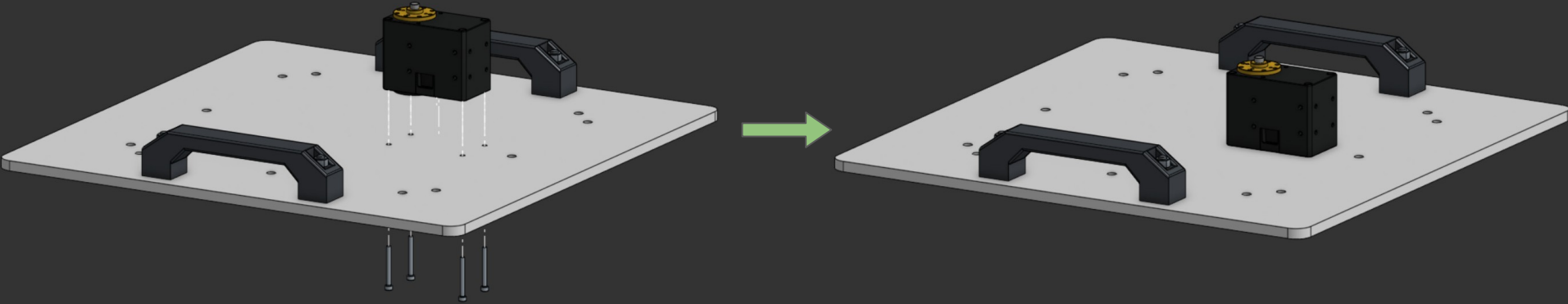


Parts

- Robotis XM540-W270-T Actuator
- 4x Robotis Spacers

Step 11

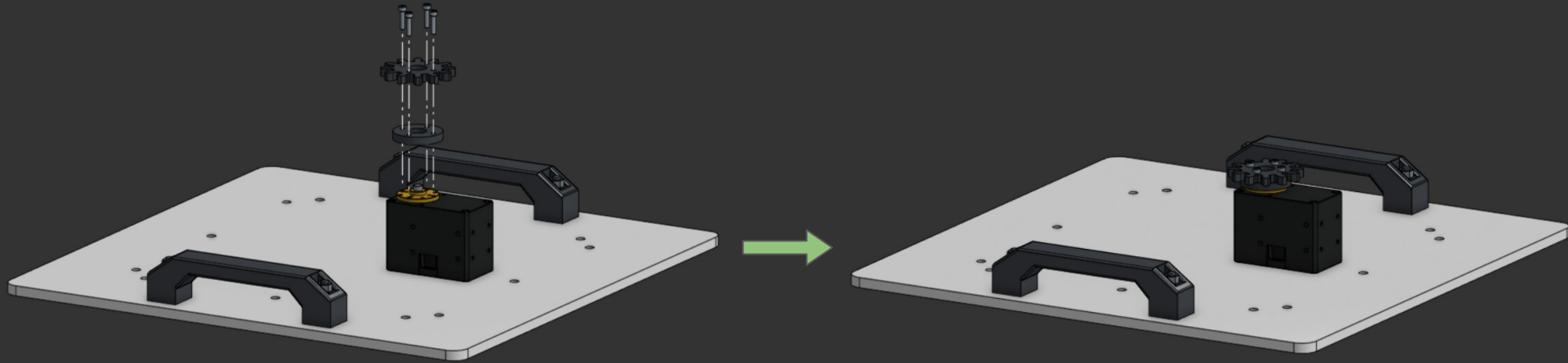
Bolt the back of the actuator to the Base Plate



Parts

- Previous step
- Base Plate (with handles)
- 4x M2.5x25mm Socket Head Bolt

Step 12 Bolt the 12T Gear on top of the Gear Spacer to the actuator

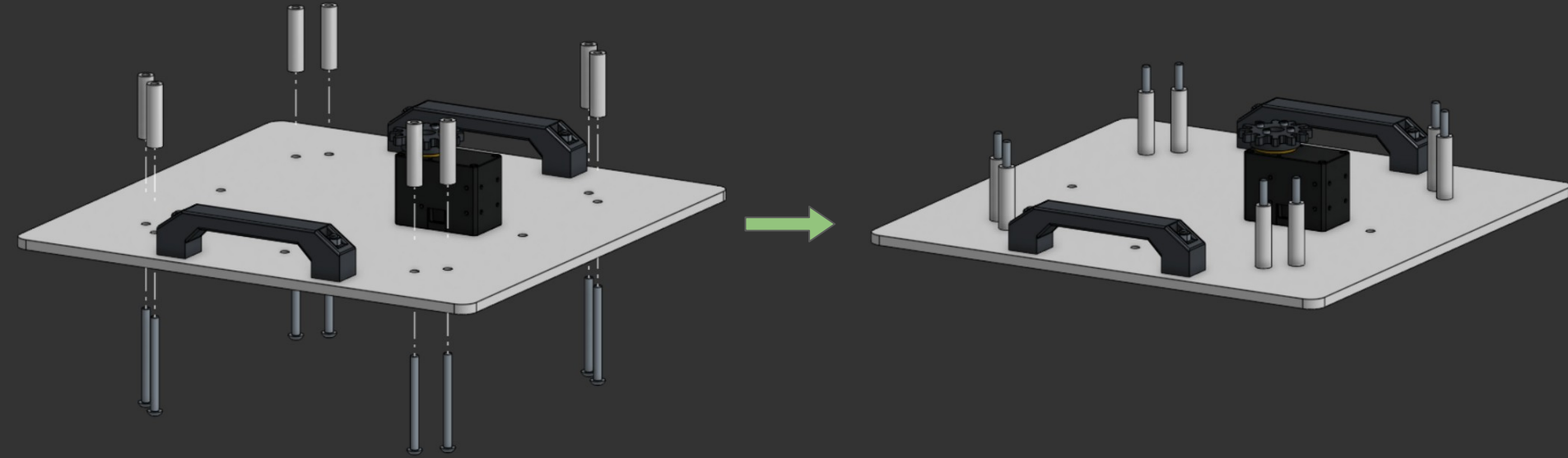


Parts
-Previous step
-12T Gear
-Gear Spacer

-4x M2.5x12mm Socket Head Bolt

Step 13

Insert M5x60mm Button Head Bolts through eight corner holes, and place 40mm spacers on top

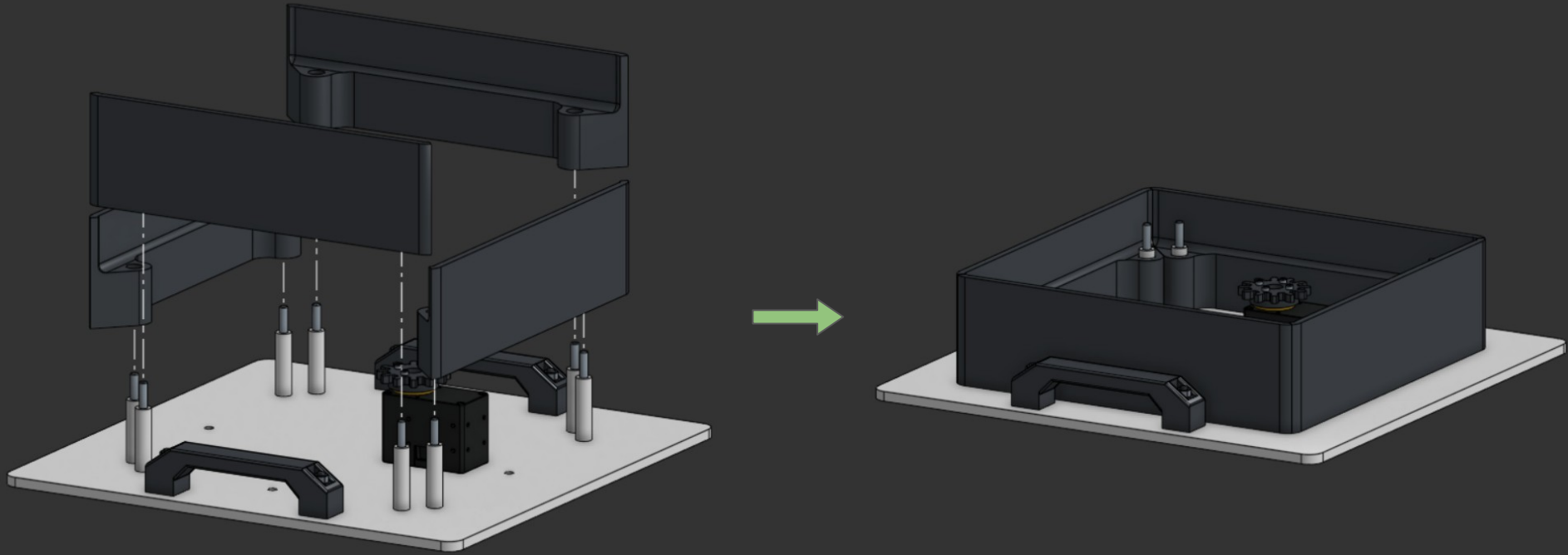


Parts

- Previous step
- 8x M5*60mm Button Head Bolt
- 8x 5x10x40mm M5 Threaded Spacers

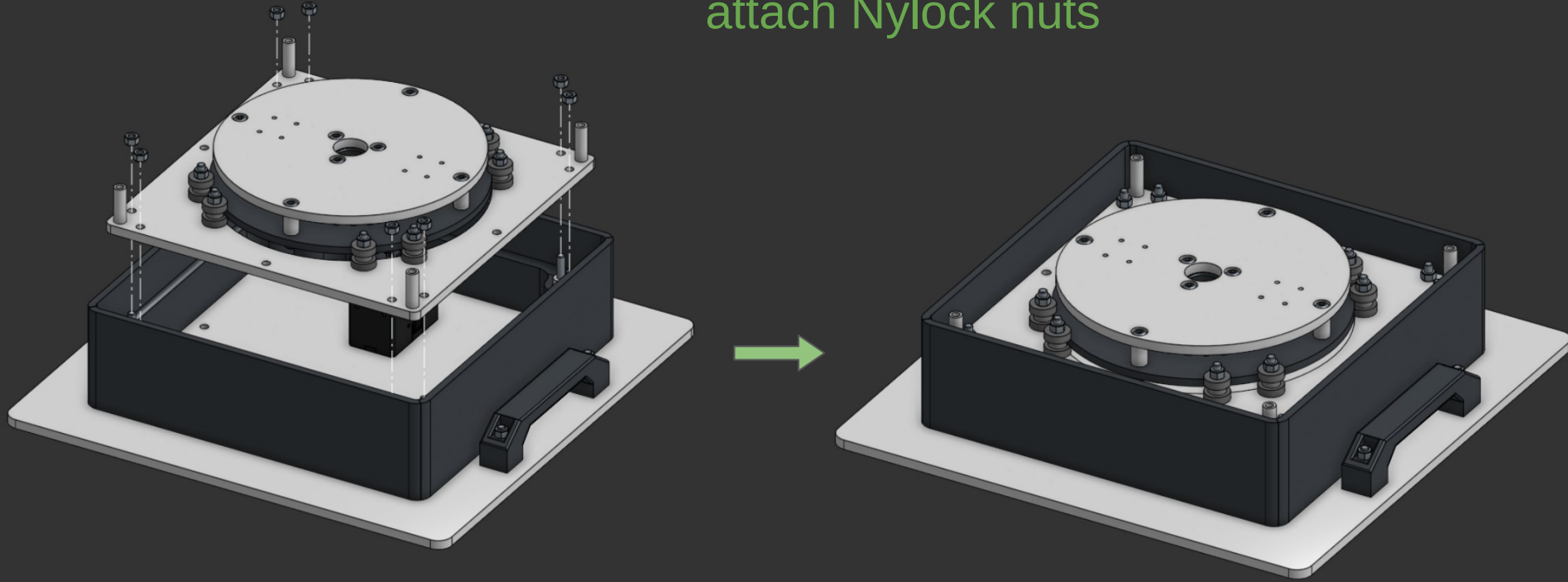
Step 14

Insert Base Walls onto each face of the base



Parts
-Previous step
-4x Base Wall

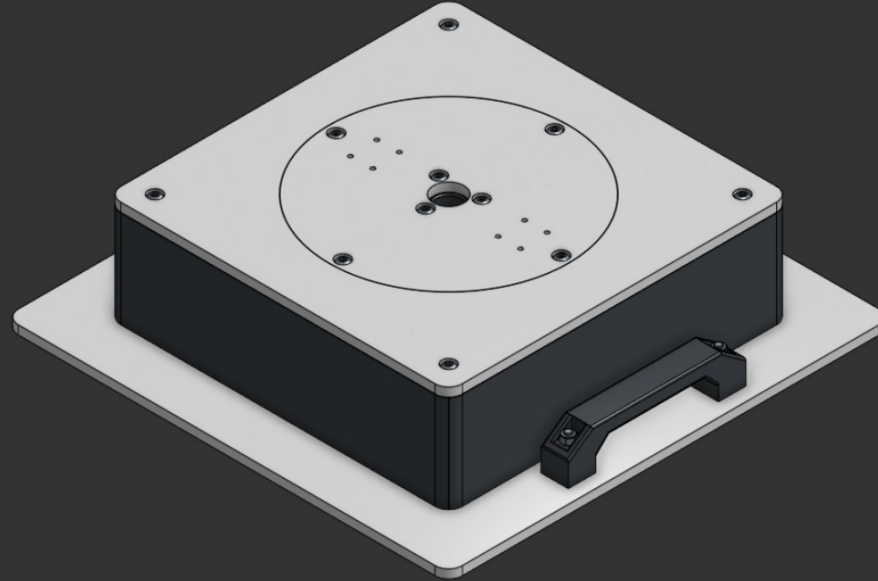
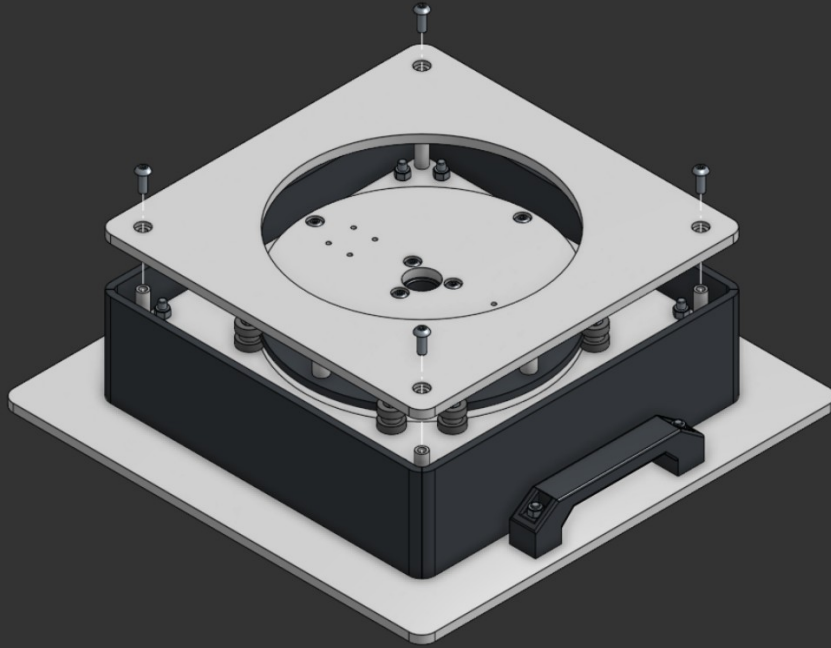
Step 15 Insert the Arm Base Plate Assembly on top of the bolts and attach Nylock nuts



Parts
-Previous step
-Arm Structure Plate and Arm Base
Plate Assembly

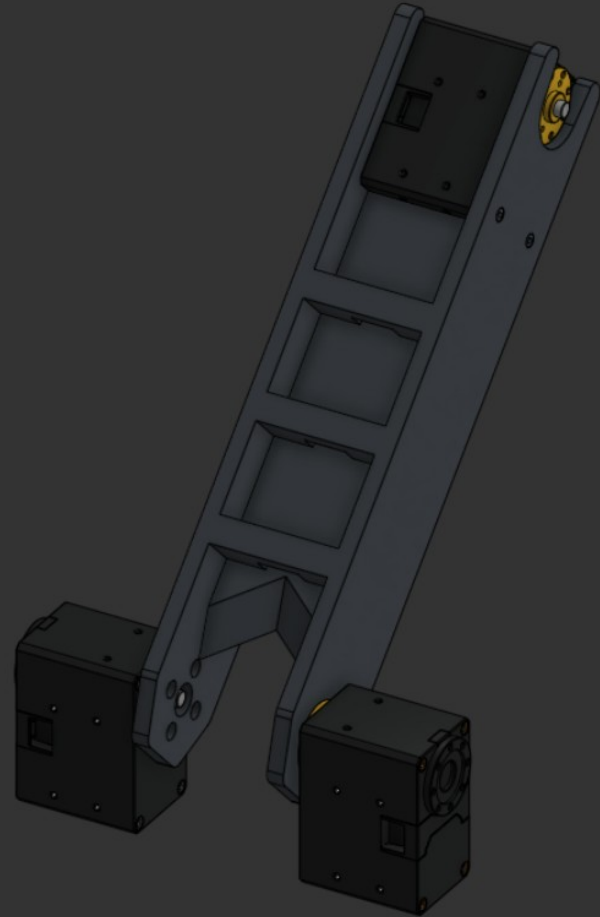
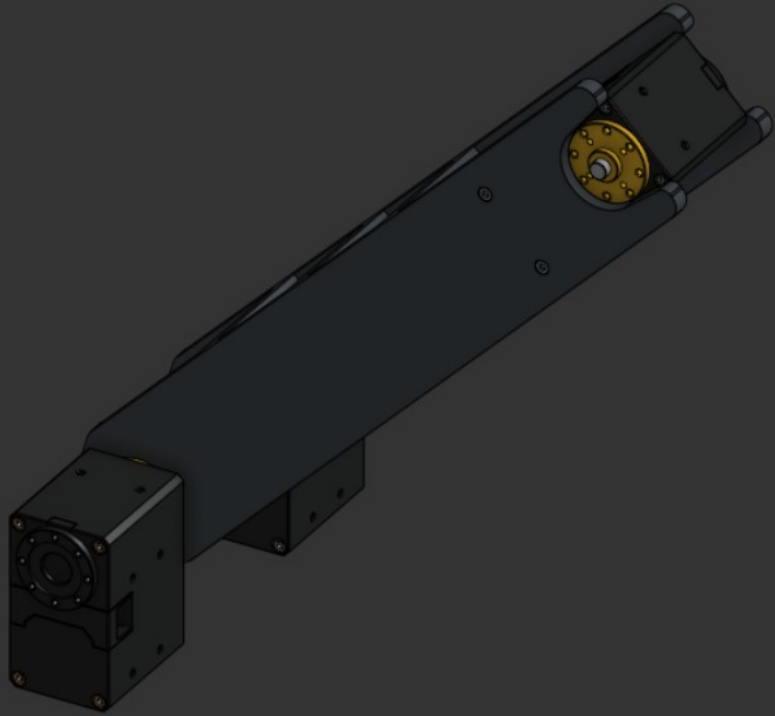
-8x Nylock Nuts

Step 16 Place the Base Top Plate around the Arm Base Plate and bolt each corner



Parts
-Previous step
-4x M5x14mm Button Head Bolt

Arm Shoulder Assembly Instructions



Parts

Major Components:

Arm Shoulder Piece

3x Robotis XM540-W270-T

Actuator

Minor Components:

8x M2.5x6mm Bolts

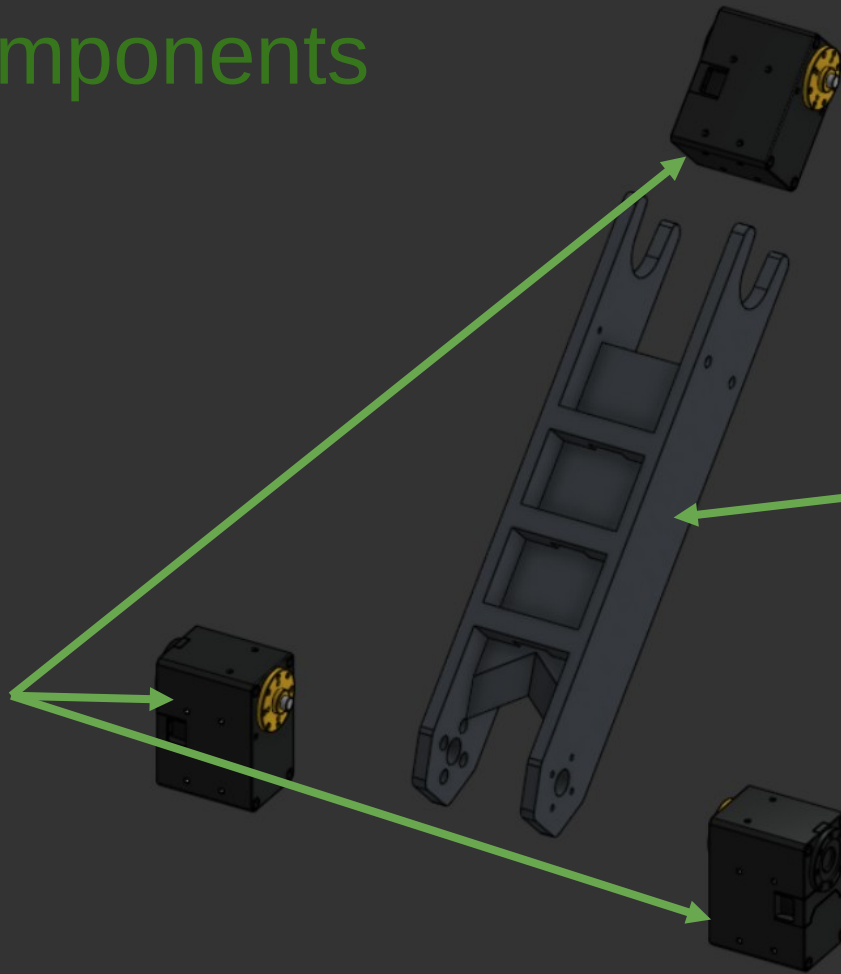
4x M2.5x25mm Bolts

4x Robotis Spacers

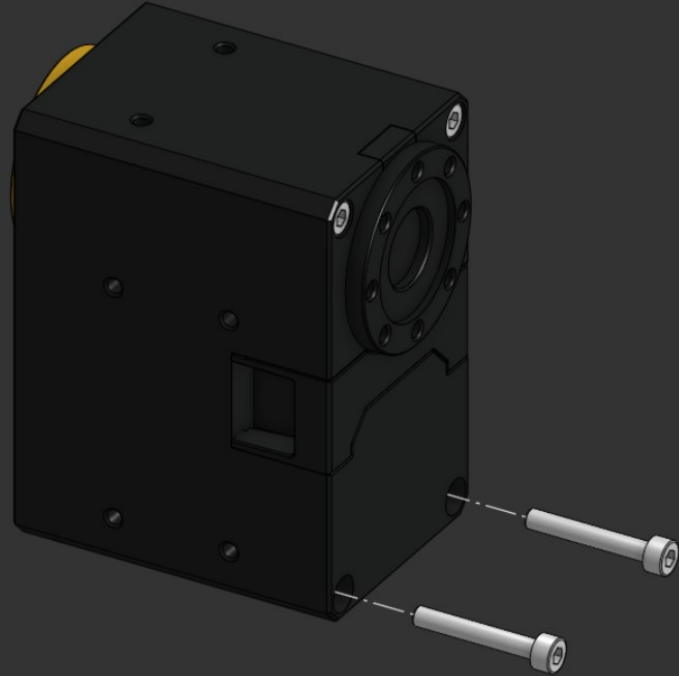
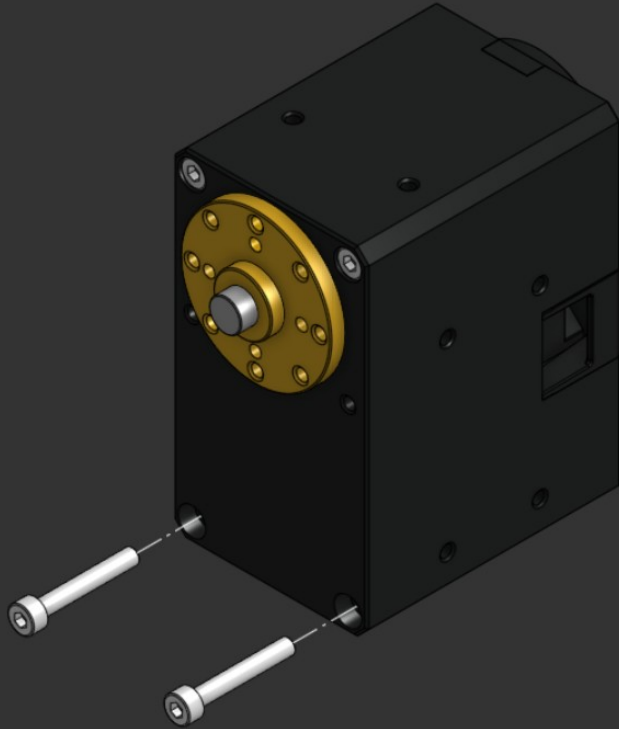
Major Components

Robotis
XM540-W270-T
Motors

Arm Shoulder
Piece



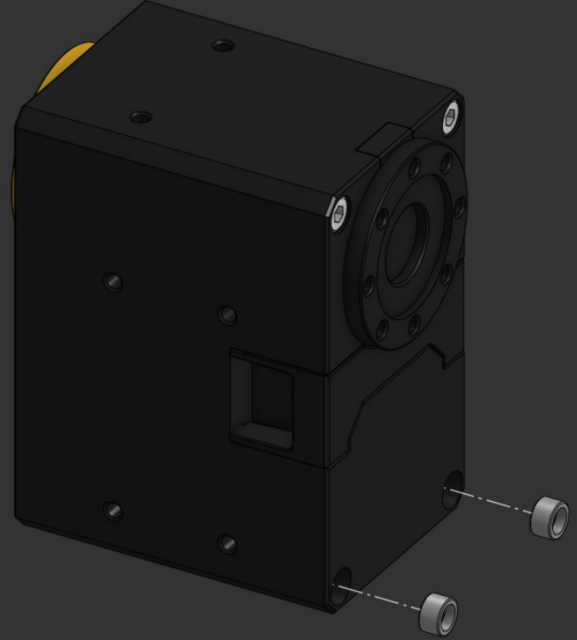
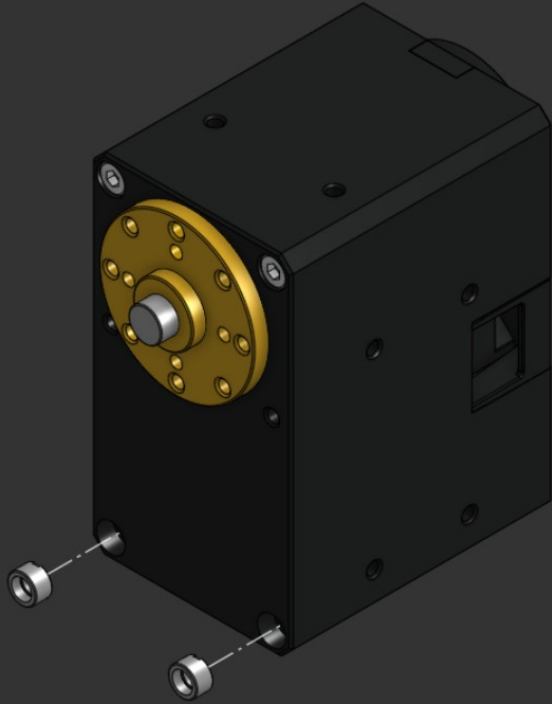
Step 1 Remove the bottom two bolts from both sides of the actuator



Parts
-Robotis XM540-W270-T Actuator

Step 2

Insert spacer rings into the actuator

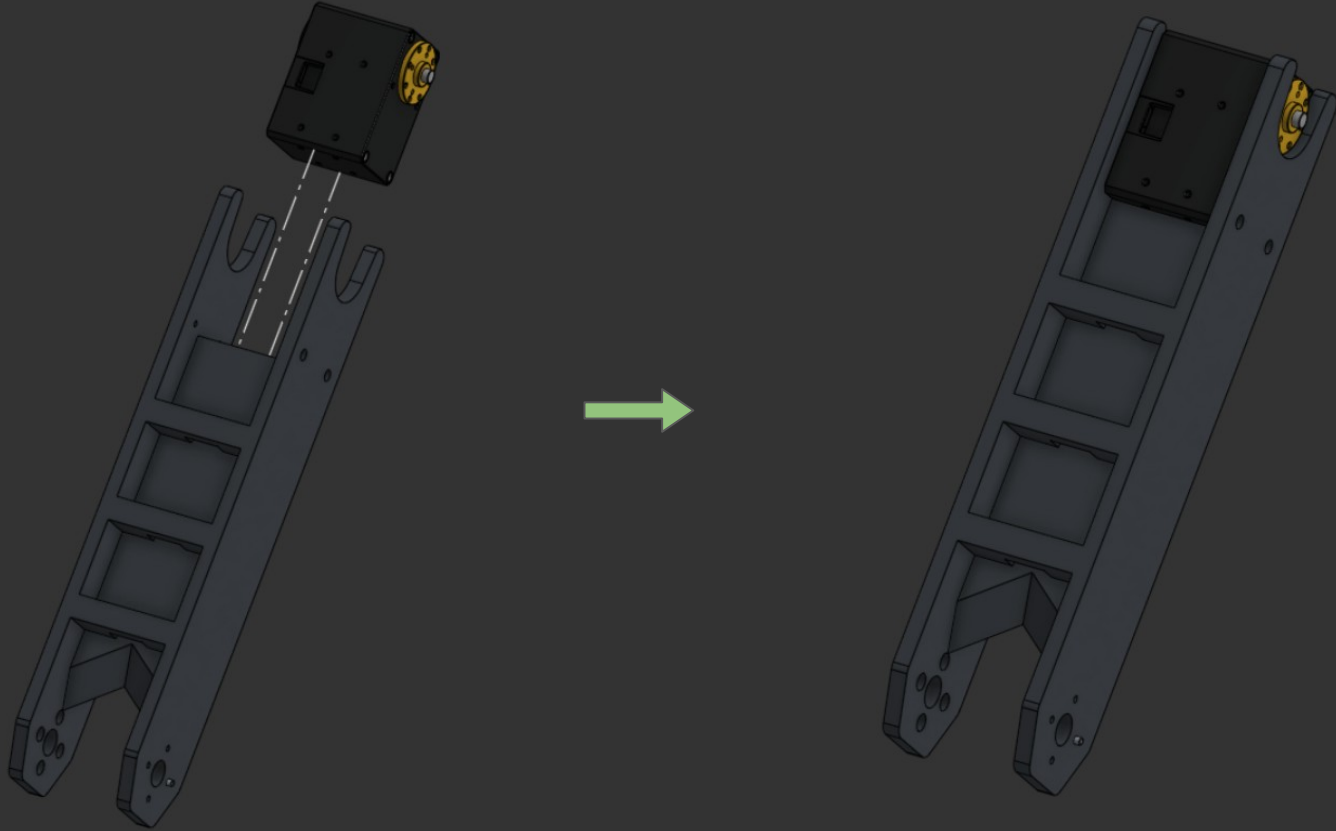


Parts

- Robotis XM540-W270-T Actuator
- 4x Robotis Spacers

Step 3

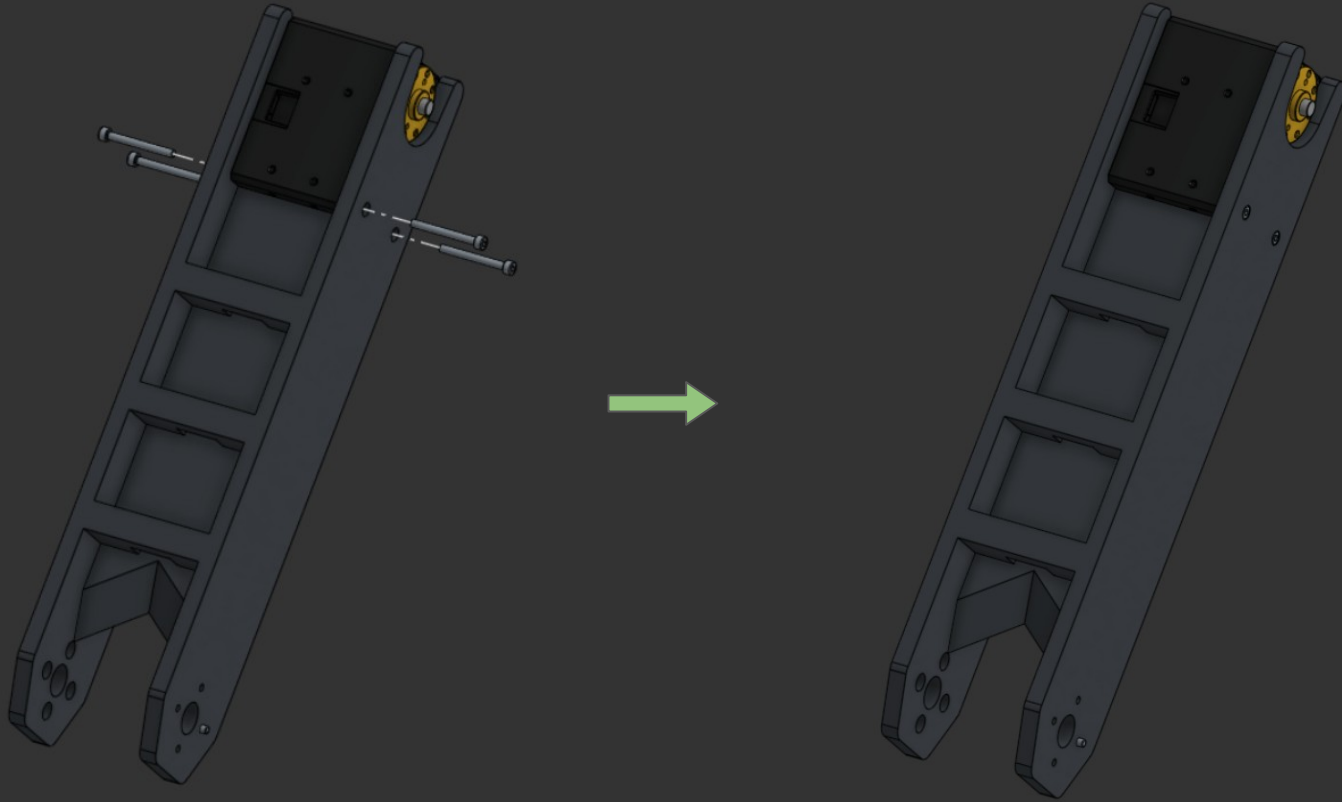
Place the actuator in the top of the shoulder piece



Parts
-Previous step
-Arm Shoulder Piece

Step 4

Bolt the actuator into the top of the shoulder piece



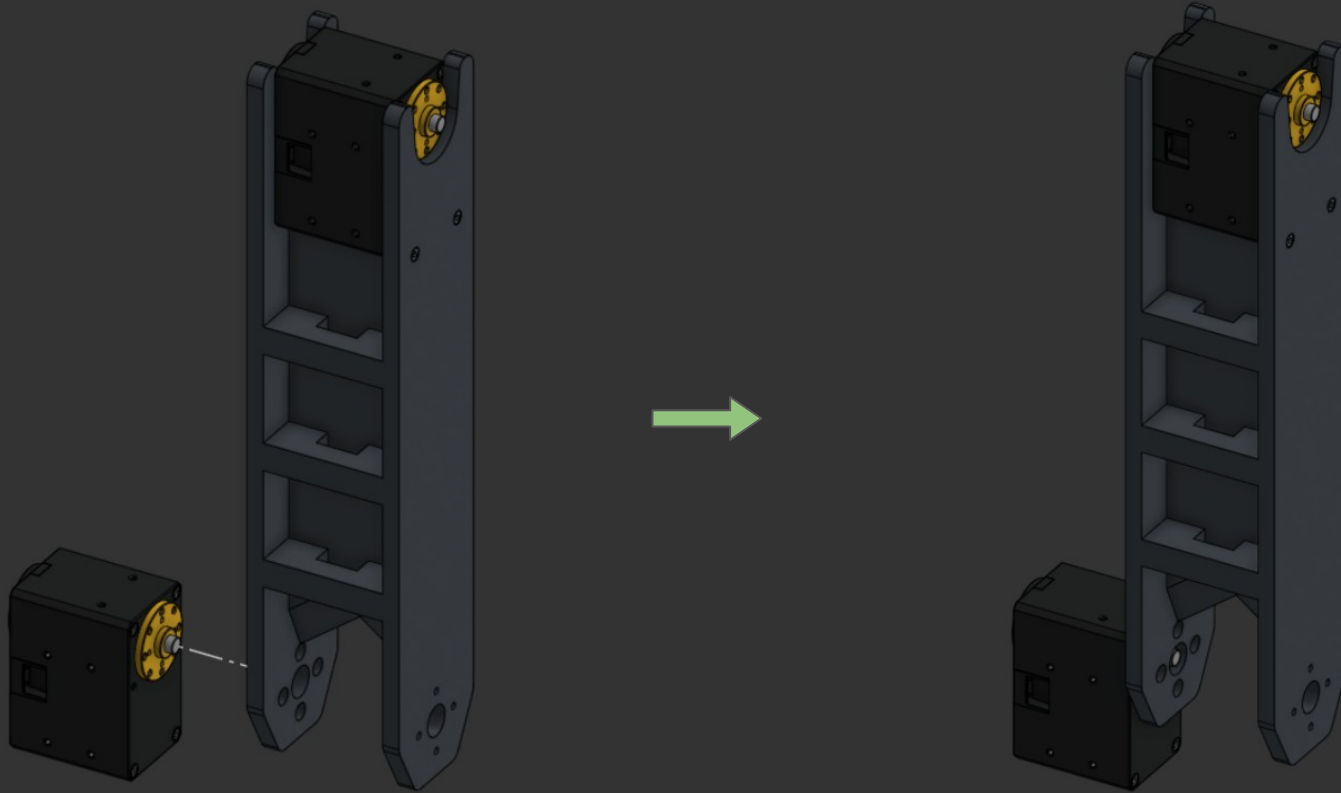
Parts

-Previous step

-4x M2.5x25mm Socket Head Bolt

Step 5

Align the actuator with the bottom of the shoulder piece



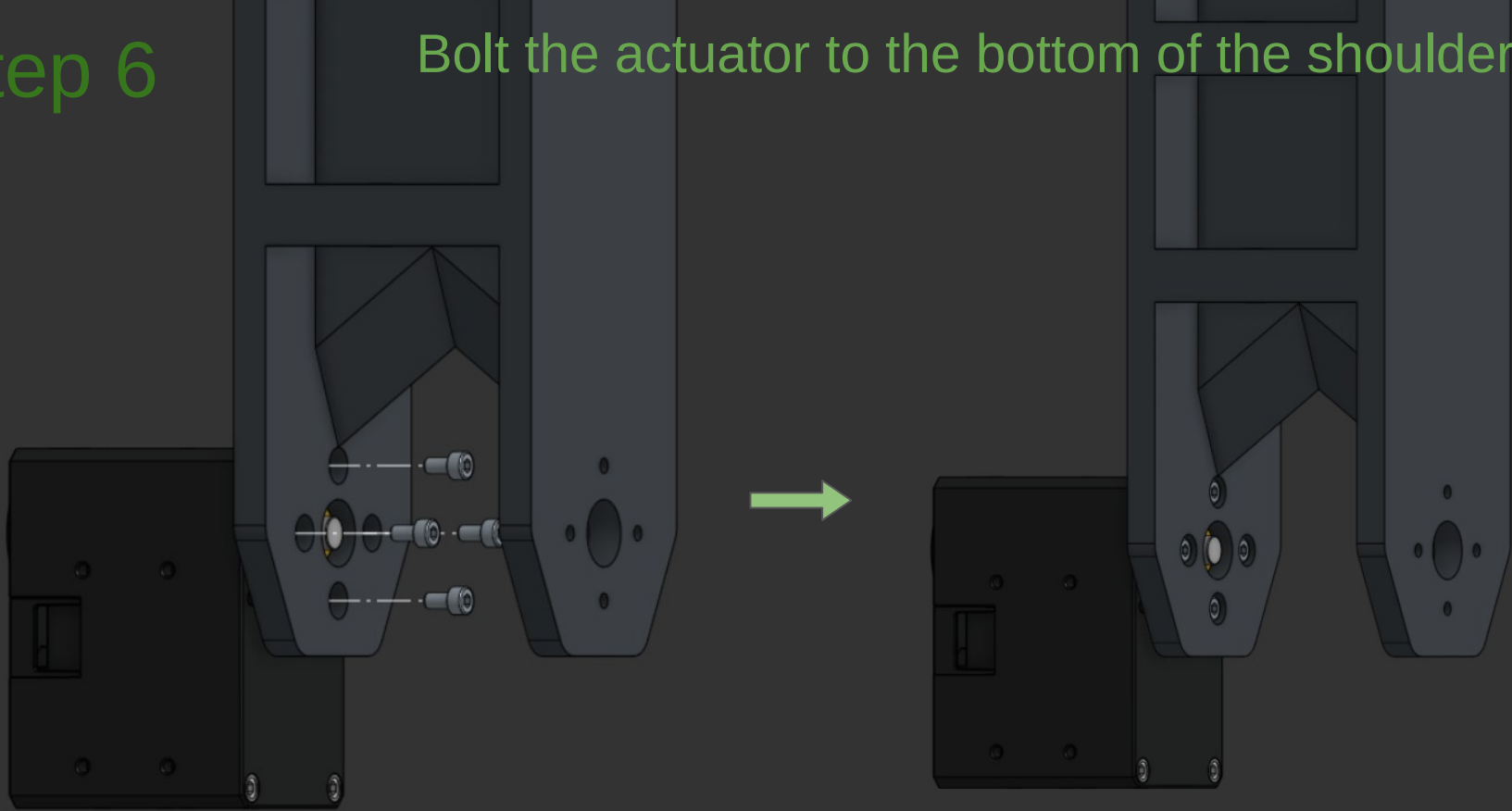
Parts

-Previous step

-Robotis XM540-W270-T Actuator

Step 6

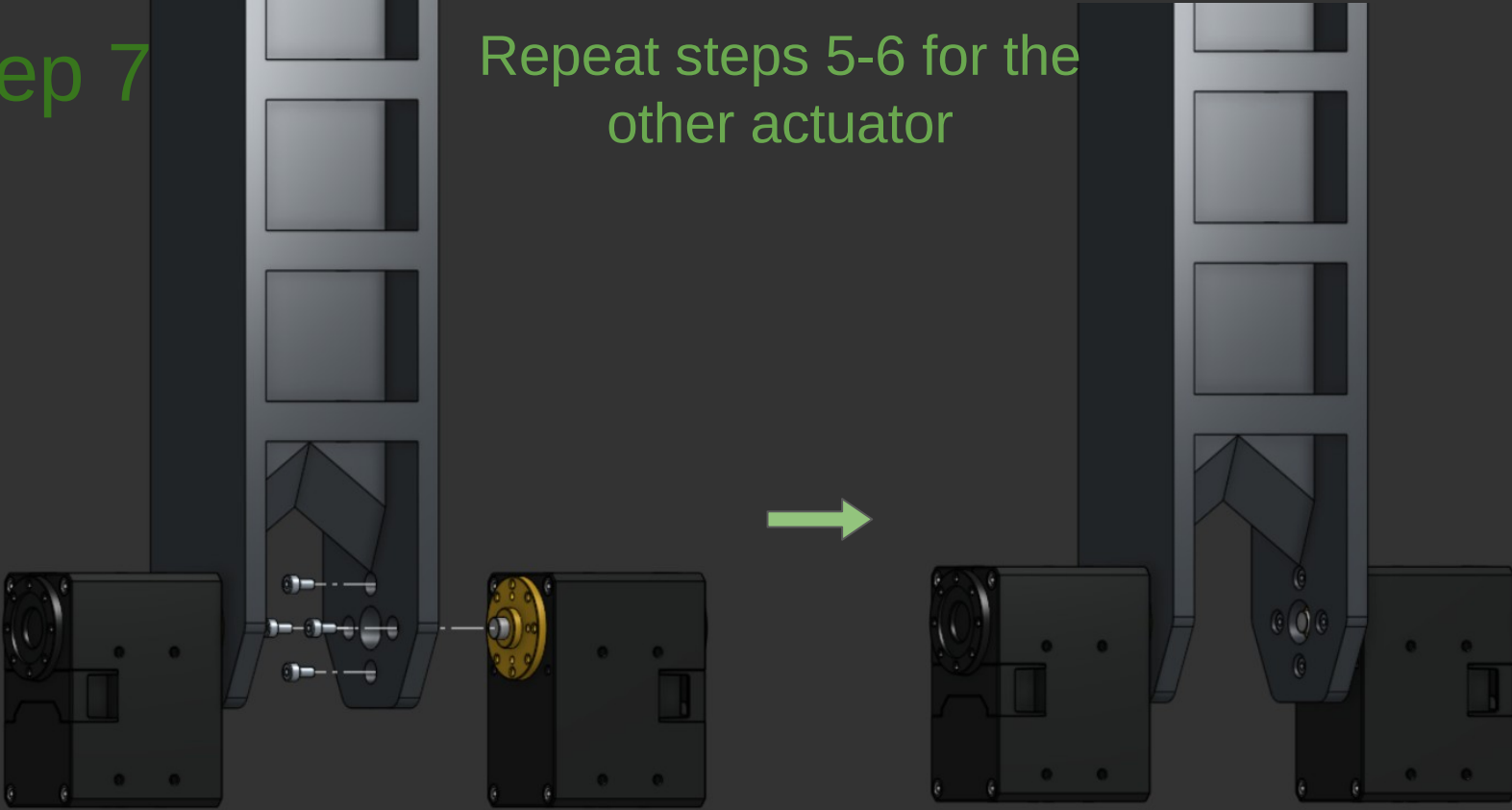
Bolt the actuator to the bottom of the shoulder piece



Parts
-Previous step
-4x M2.5x6mm Bolts

Step 7

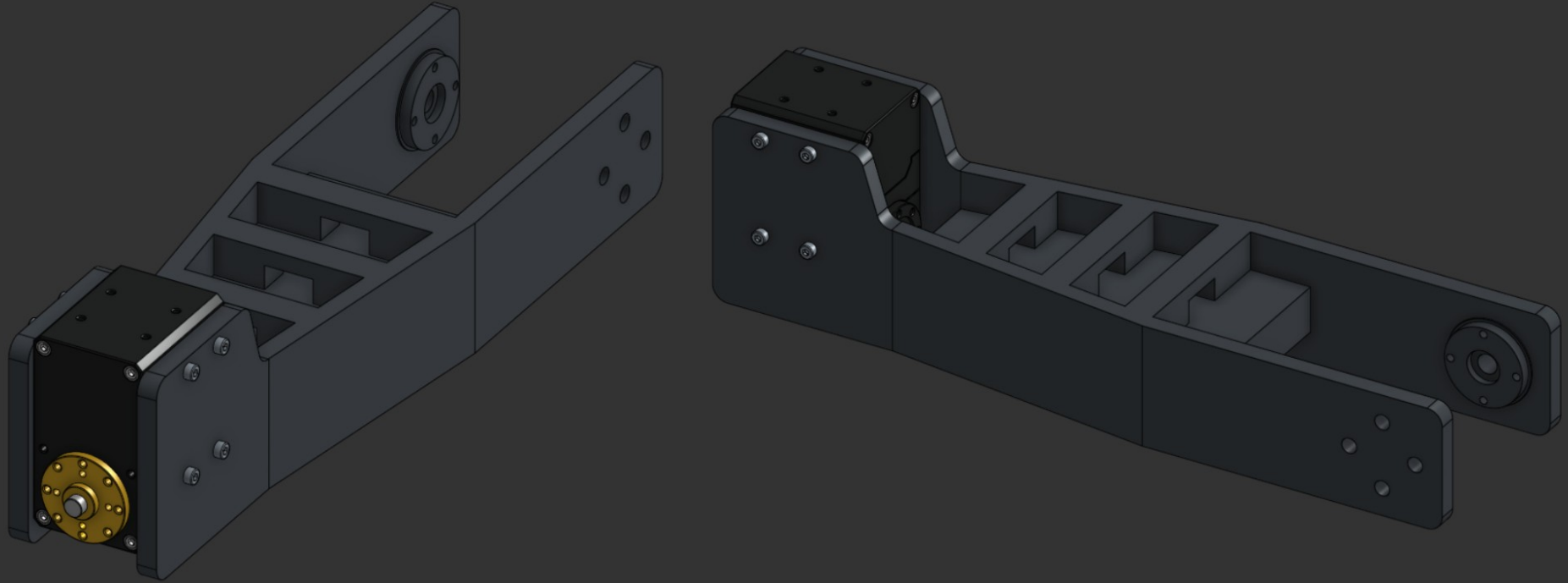
Repeat steps 5-6 for the other actuator



Parts

- Previous step
- Robotis XM540-W270-T Actuator
- 4x M2.5x6mm Bolts

Arm Elbow Assembly Instructions



Parts

Major Components:

Arm Elbow Piece

1x Robotis XM540-W270-T

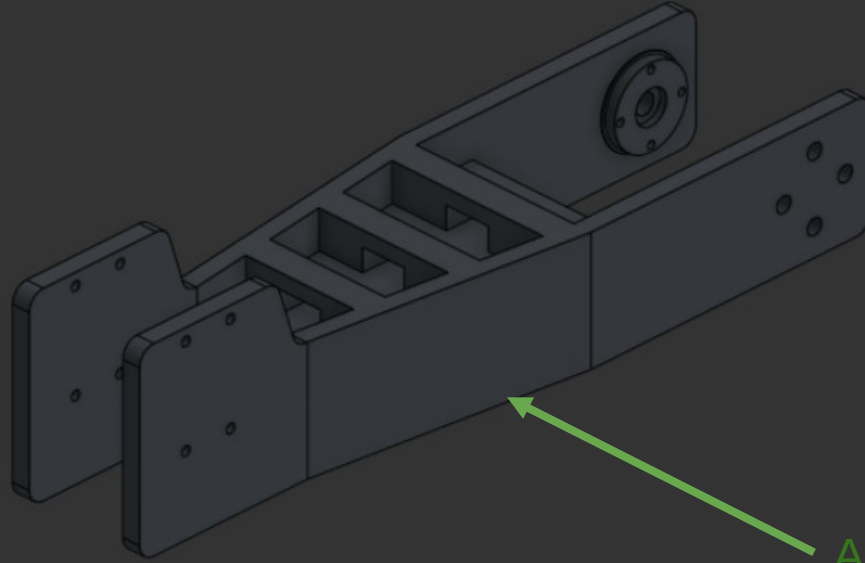
Actuator

Minor Components:

8x M2.5x8mm Socket Head Bolt

Major Components

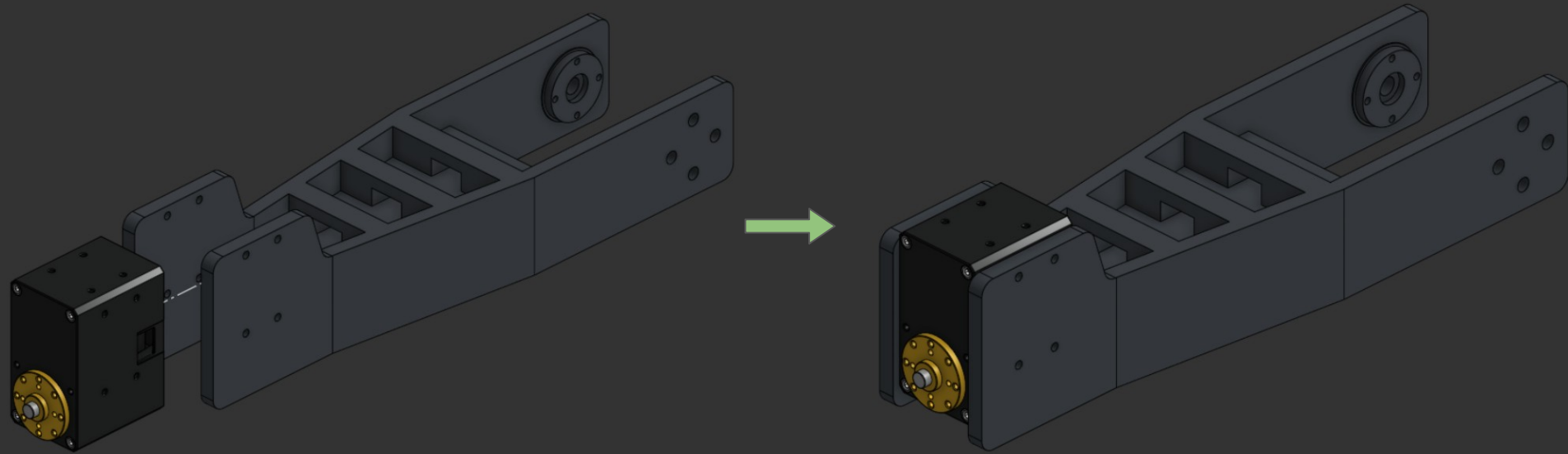
Robotis
XM540-W270-T
Actuator



Arm Elbow Piece

Step 1

Insert the actuator between the ends of the elbow piece

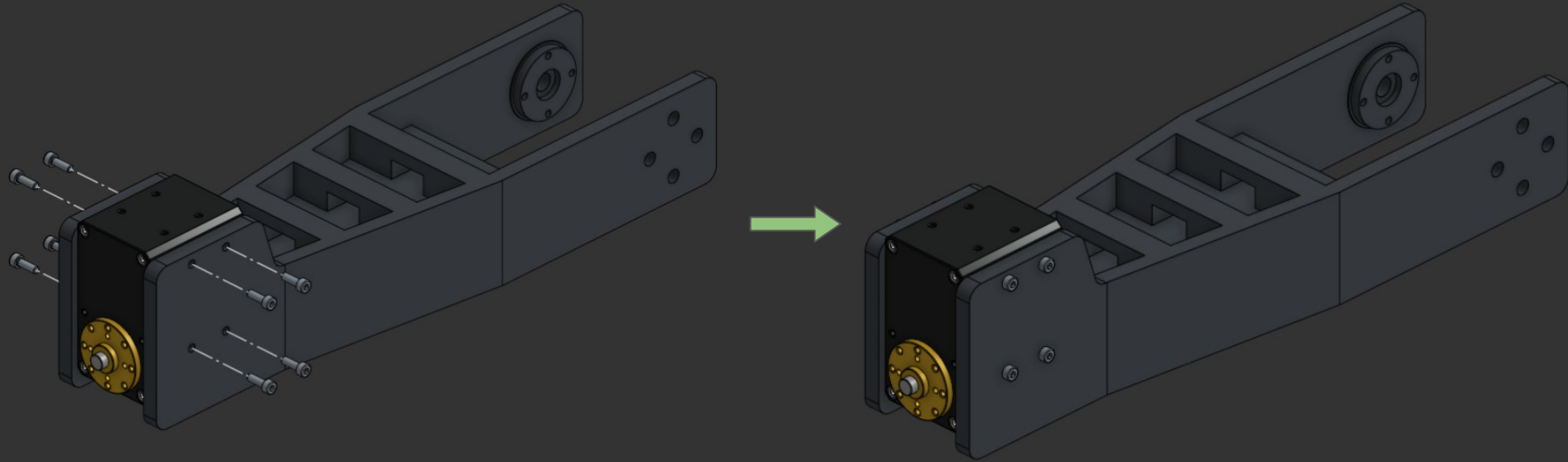


Parts

- Arm Elbow Piece
- Robotis XM540-W270-T Actuator

Step 2

Bolt the actuator into the ends of the elbow piece

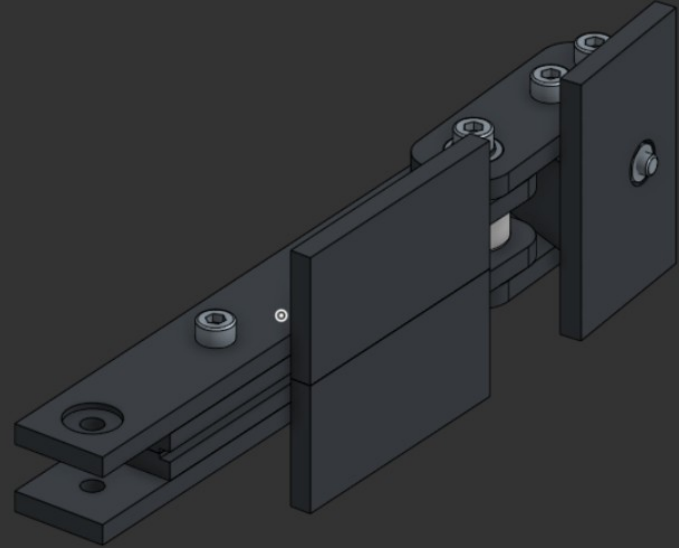
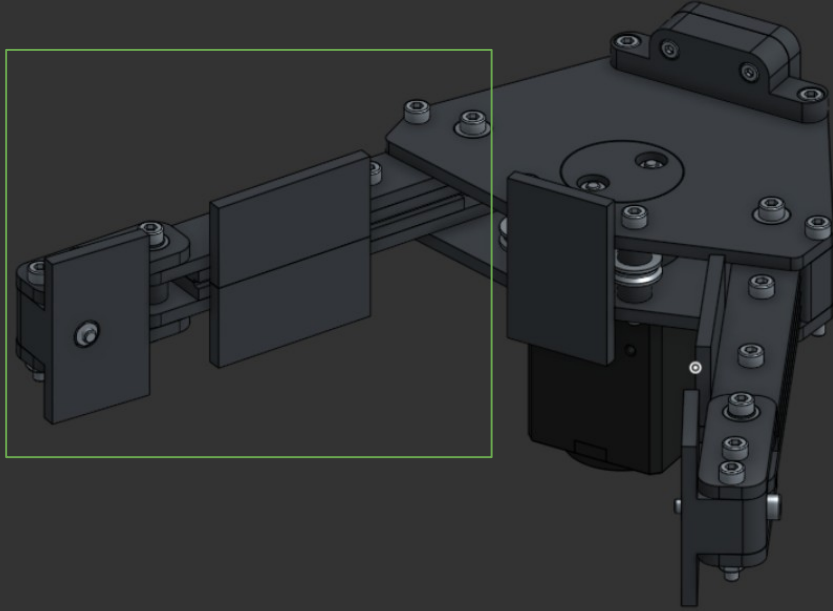


Parts

-Previous step

-4x M2.5x8mm Socket Head Bolt

Arm Gripper Finger Assembly Instructions



Parts

Major Components:

- 2x Spring
- 2x Spring Spacer
- 4x Finger Front Plate
- 4x Finger Center Plate
- 2x Front Finger Pad
- 4x Center Finger Pad (two top pieces, two bottom pieces)

Minor Components:

- 6x M3x25mm Socket Head Bolt
- 4x M3x20mm Socket Head Bolt
- 4x 3x8x4mm Bearing
- 10x M3 Lock Washer
- 10x M3 Nut

Major Components

Finger Front
Plate

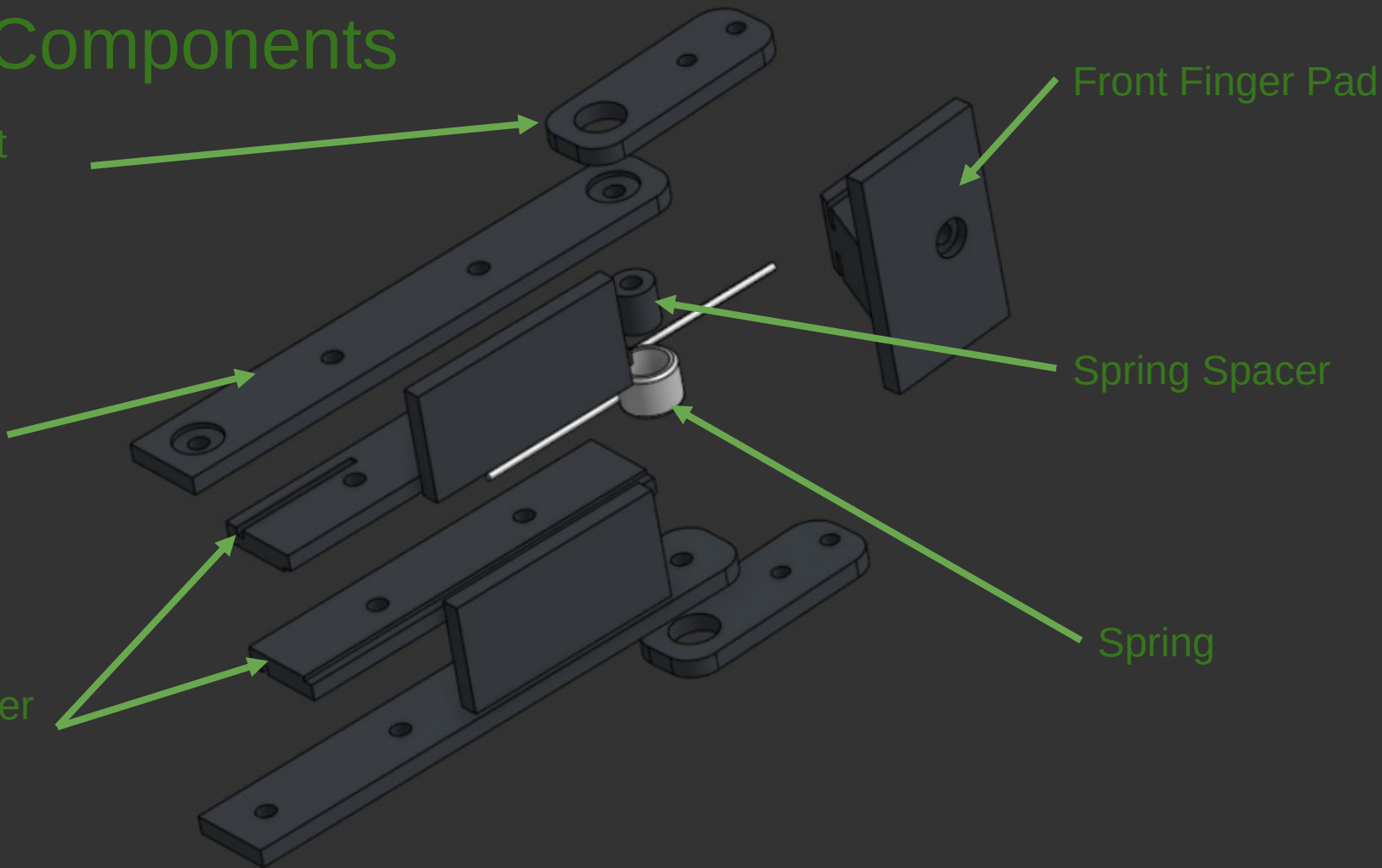
Finger
Center
Plate

Center Finger
Pads

Front Finger Pad

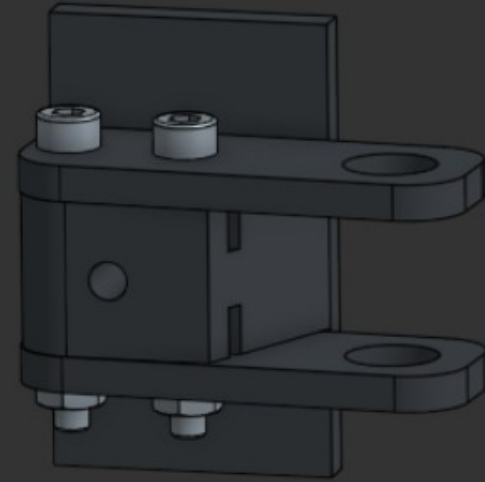
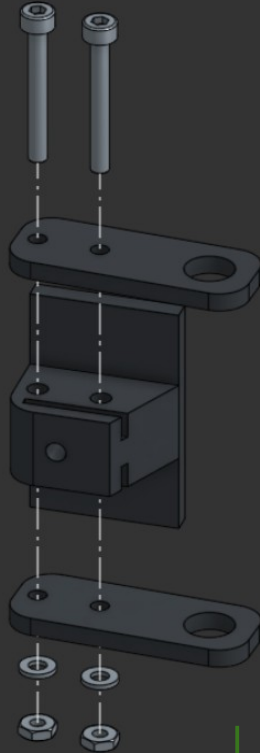
Spring Spacer

Spring



Step 1

Bolt the Front Finger Plates to the Front Finger Pad with a lock washer underneath the nut

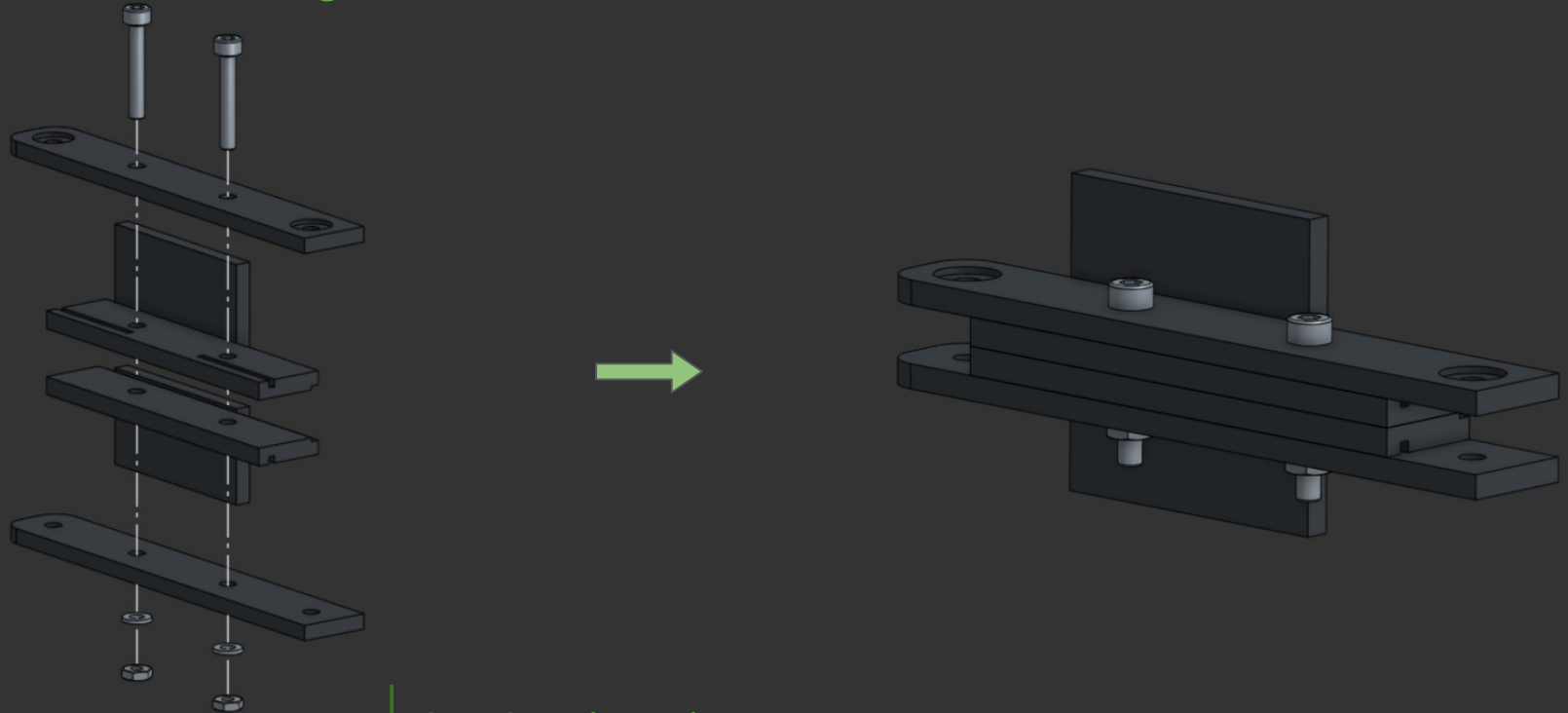


Parts
-2x Finger Front Plate
-Front Finger Pad
-2x M3x25mm Socket Head Bolt

-2x M3 Lock Washer
-2x M3 Nut

Step 2

Bolt the Center Finger Plates and Center Finger Pads together with a lock washer underneath the nut



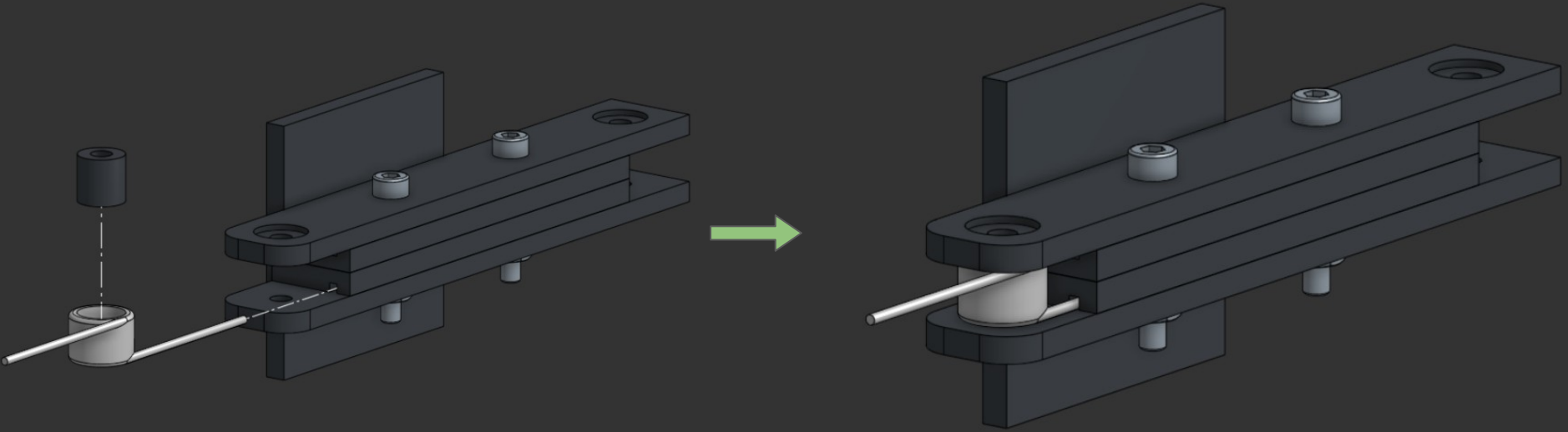
Parts

- 2x Finger Center Plate
- 2x Center Finger Pad
- 2x M3x20mm Socket Head Bolt

- 2x M3 Lock Washer
- 2x M3 Nut

Step 3

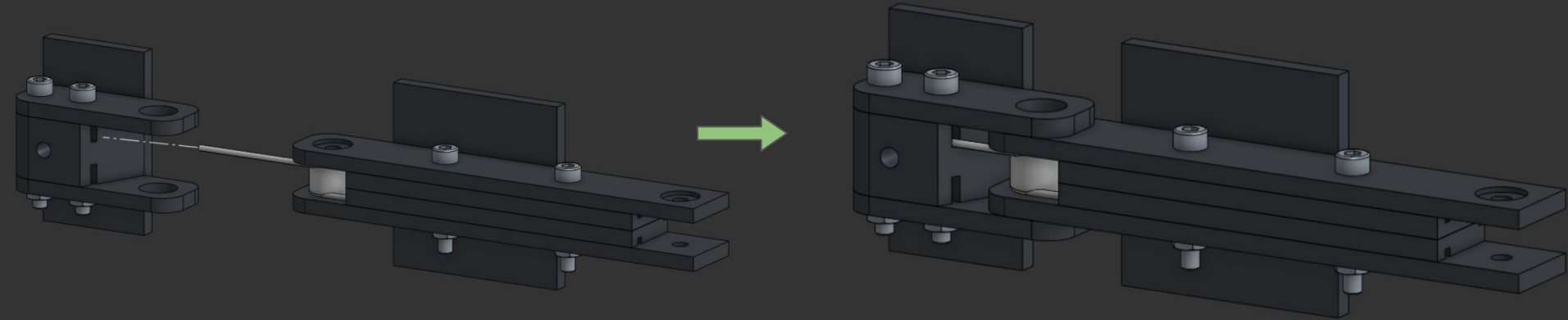
Insert the Spring Spacer into the center of the spring and then align the spring with the end of the Center Finger Plates



- Parts
- Previous step
 - Spring
 - Spring Spacer

Step 4

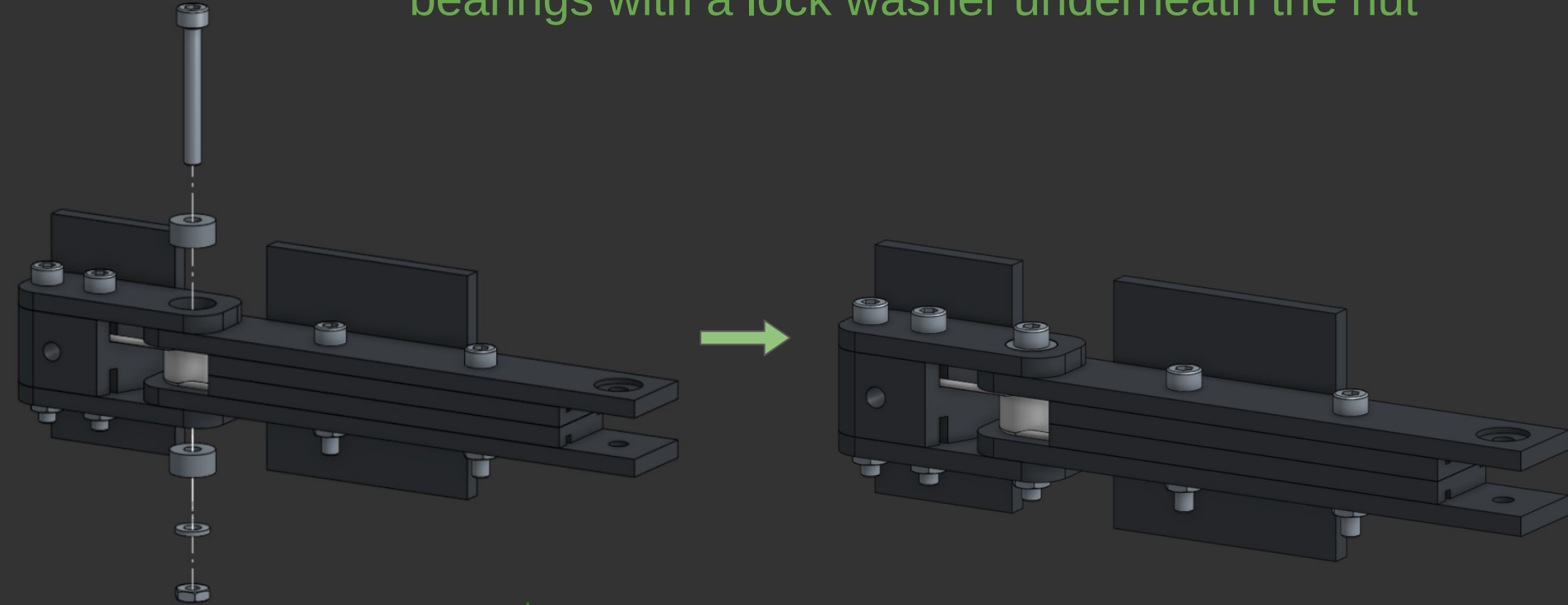
Align the front section of the finger with the center section



- Parts
- Previous step
 - Finger Front Assembly

Step 5

Bolt the front and center sections together on top of the bearings with a lock washer underneath the nut

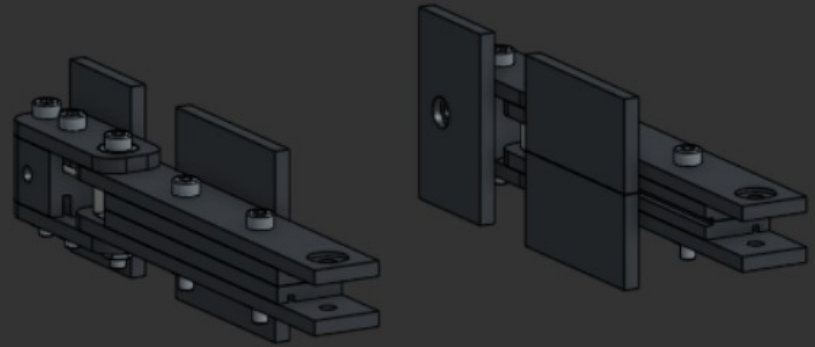
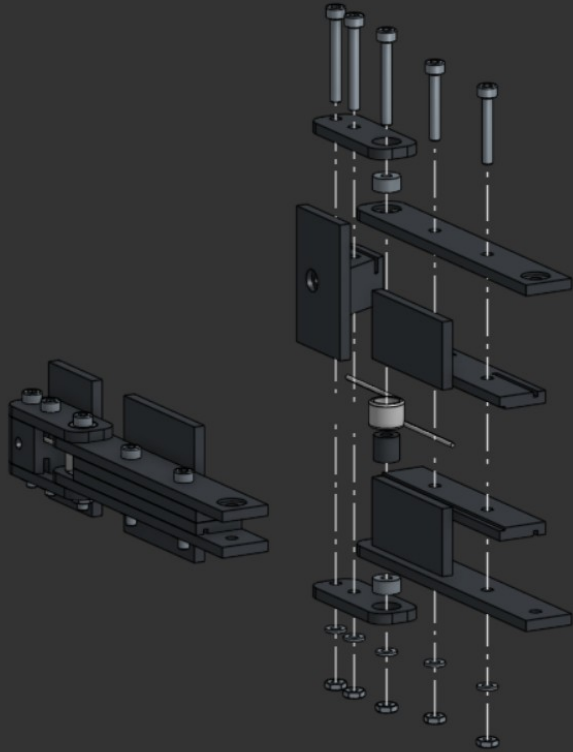


Parts
-Previous step
-M3x25mm Socket Head Bolt
-2x 3x8x4mm Bearing

-M3 Lock Washer
-M3 Nut

Step 6

Repeat steps 1-5 for a mirror finger

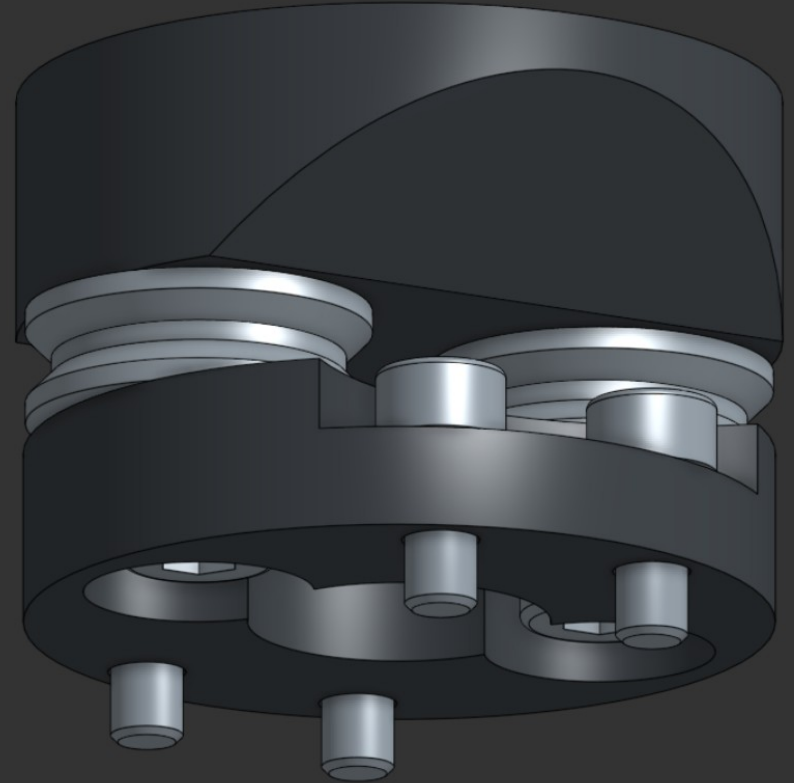
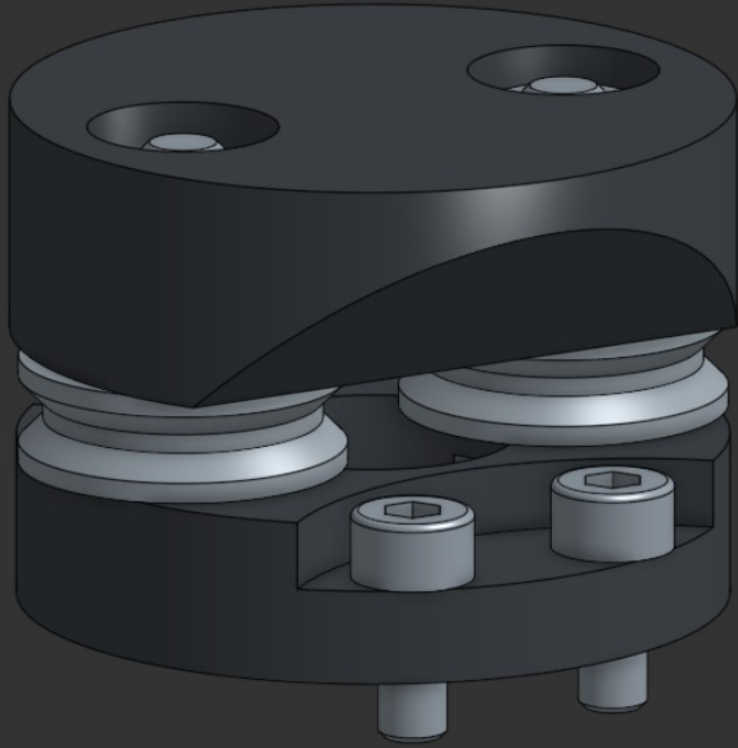


Parts
-Previous step
-Front Finger Pad
-2x Finger Front Plate

-2x Finger Center Pads
-2x Finger Center Plate
-Spring
-Spring Spacer

-2x 3x8x4mm Bearing
-2x M3x20mm Socket Head Bolt
-3x M3x25mm Socket Head Bolt
-2x M3 Lock Washer
-2x M3 Nut

Arm Gripper Coiler Assembly Instructions



Parts

Major Components:

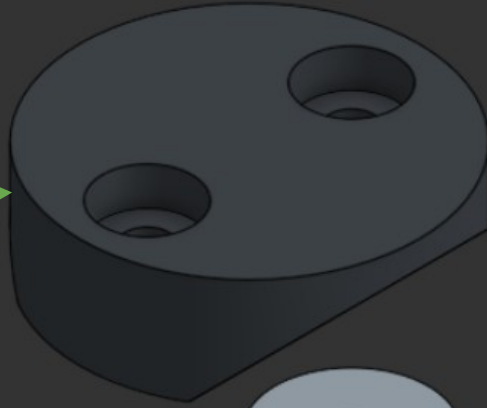
- 2x Spring
- 2x Spring Spacer
- 4x Finger Front Plate
- 4x Finger Center Plate
- 2x Front Finger Pad
- 4x Center Finger Pad (two top pieces, two bottom pieces)

Minor Components:

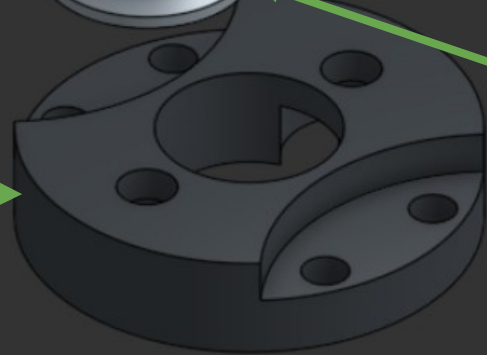
- 8x M2.5x8mm Bolt

Major Components

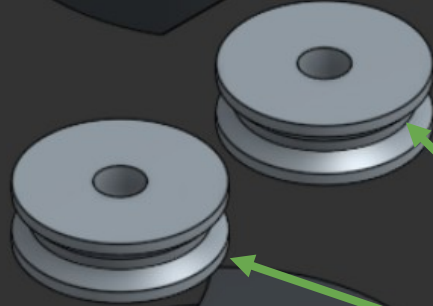
Coiler Top



Coiler
Bottom

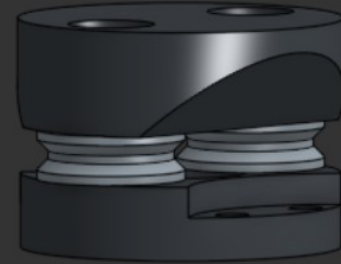
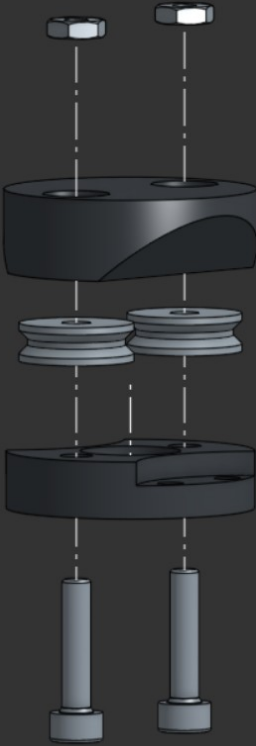


Pulleys



Step 1

Bolt the Coiler Top to the pulleys and Coiler Bottom

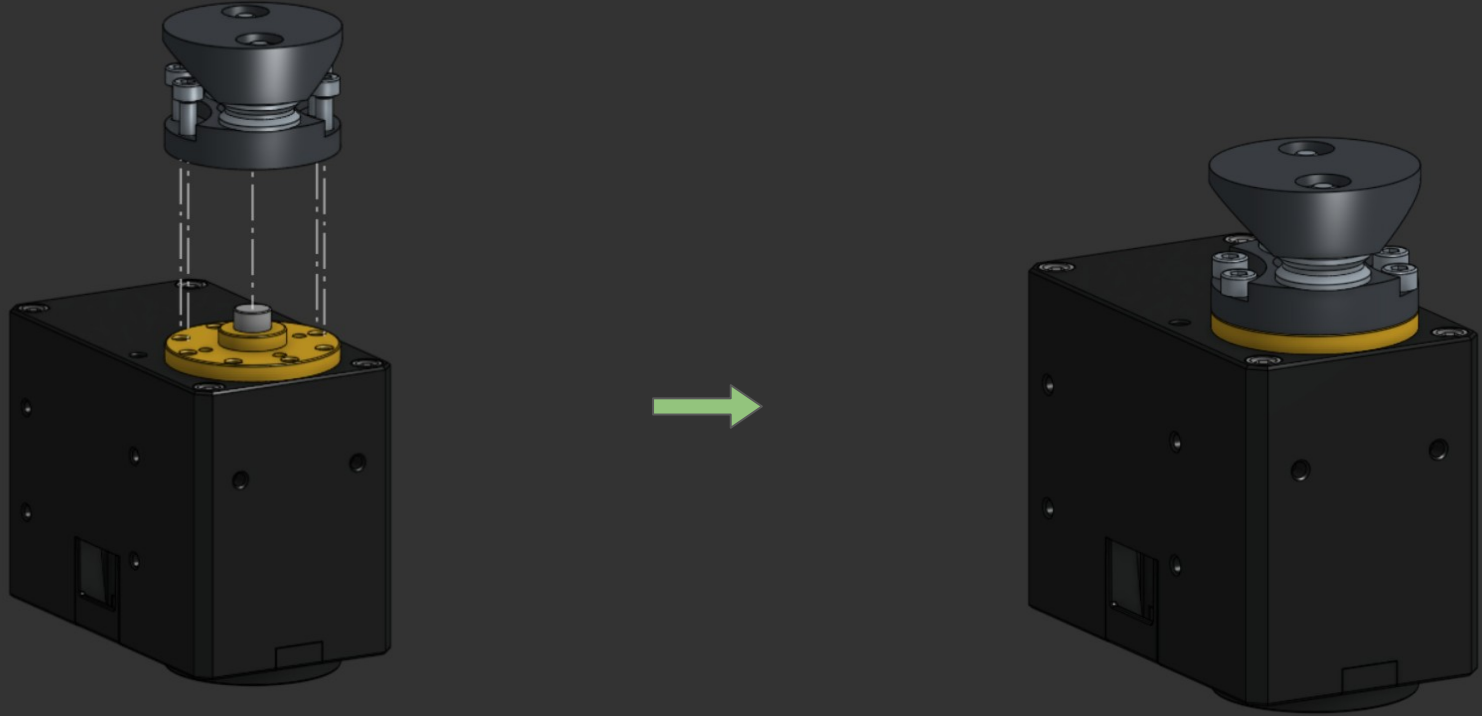


Parts
-Coiler Top
-Coiler Bottom
-2x Pulley

-2x M3x14mm Socket Head Bolt
-2x M3 Nut

Step 2

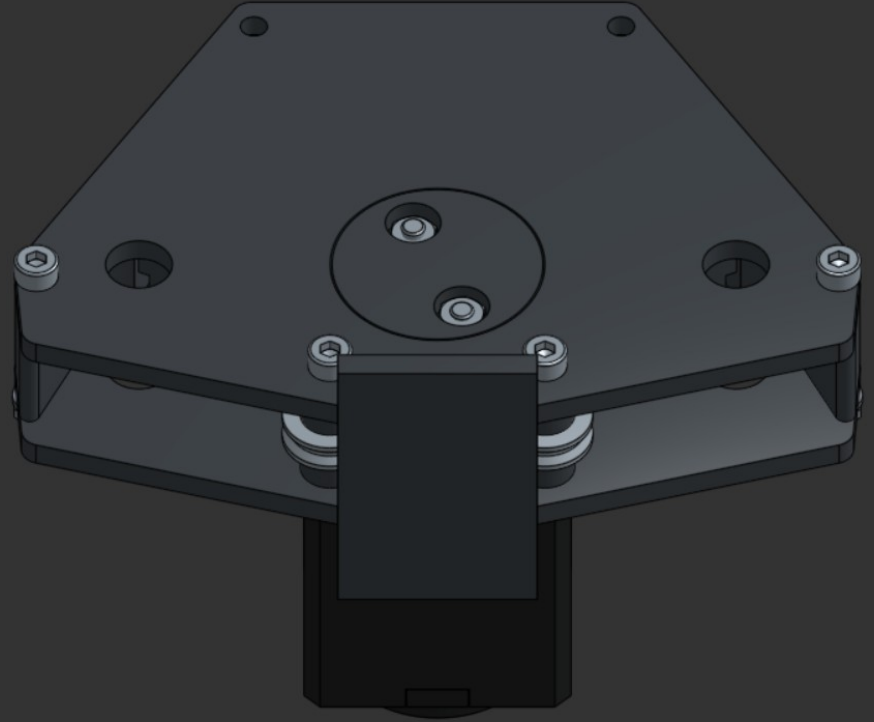
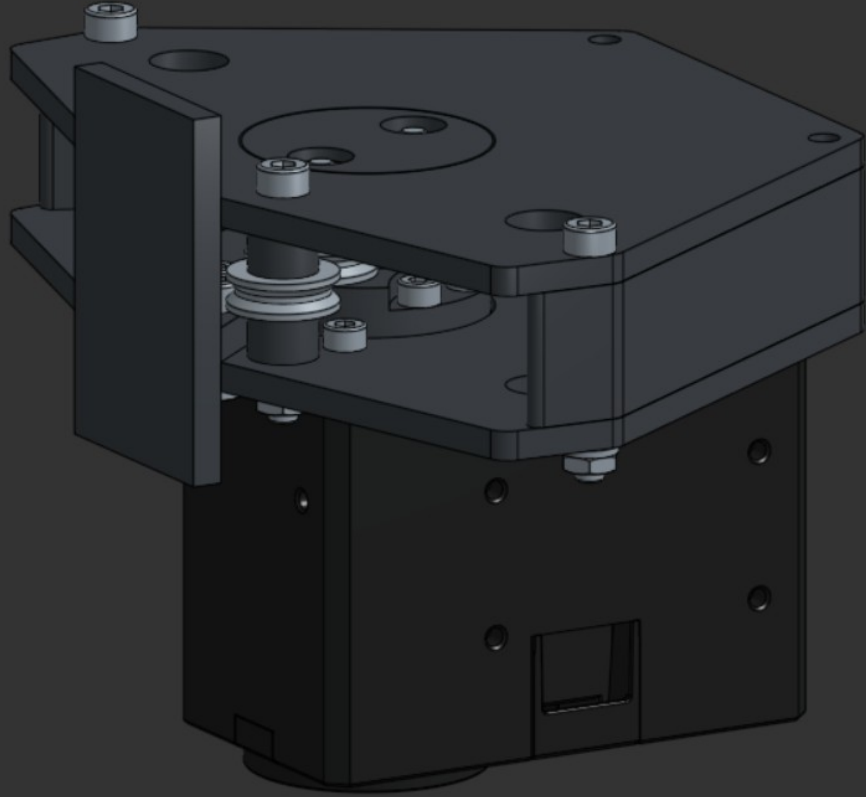
Bolt the Coiler to the horn of the actuator



Parts

- Previous step
- Robotis XM540-W270-T Actuator
- 4x M2.5x8mm Socket Head Bolt

Arm Gripper Base Assembly Instructions



Parts

Major Components:

Coiler Assembly

Gripper Base Top Plate

Gripper Base Bottom Plate

Gripper Base Front

2x Gripper Base Side Wall

2x Pulley

2x Pulley Spacer

Minor Components:

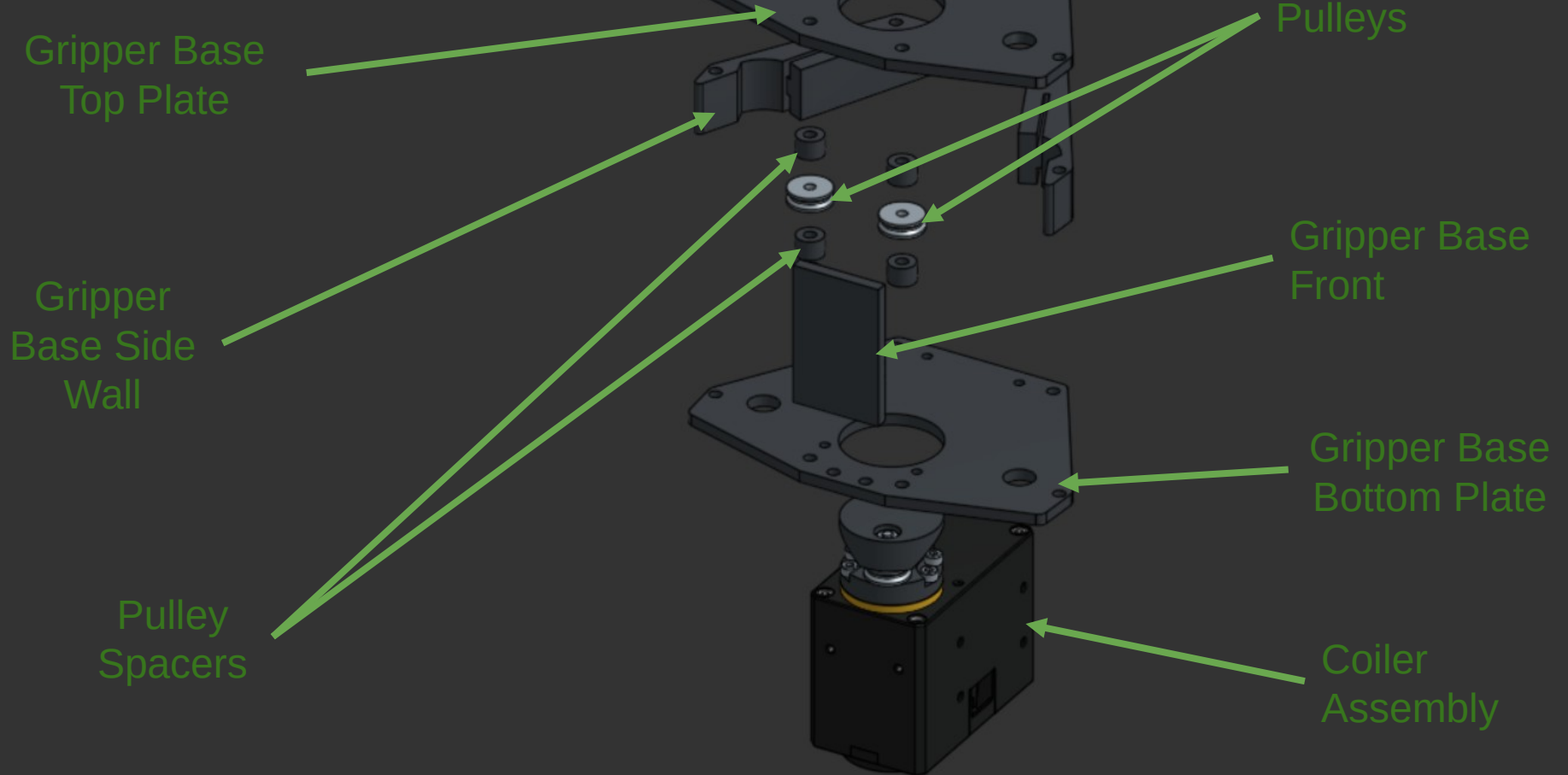
2x M3x10mm Socket Head Bolt

4x M3x25mm Socket Head Bolt

6x M3 Lock Washer

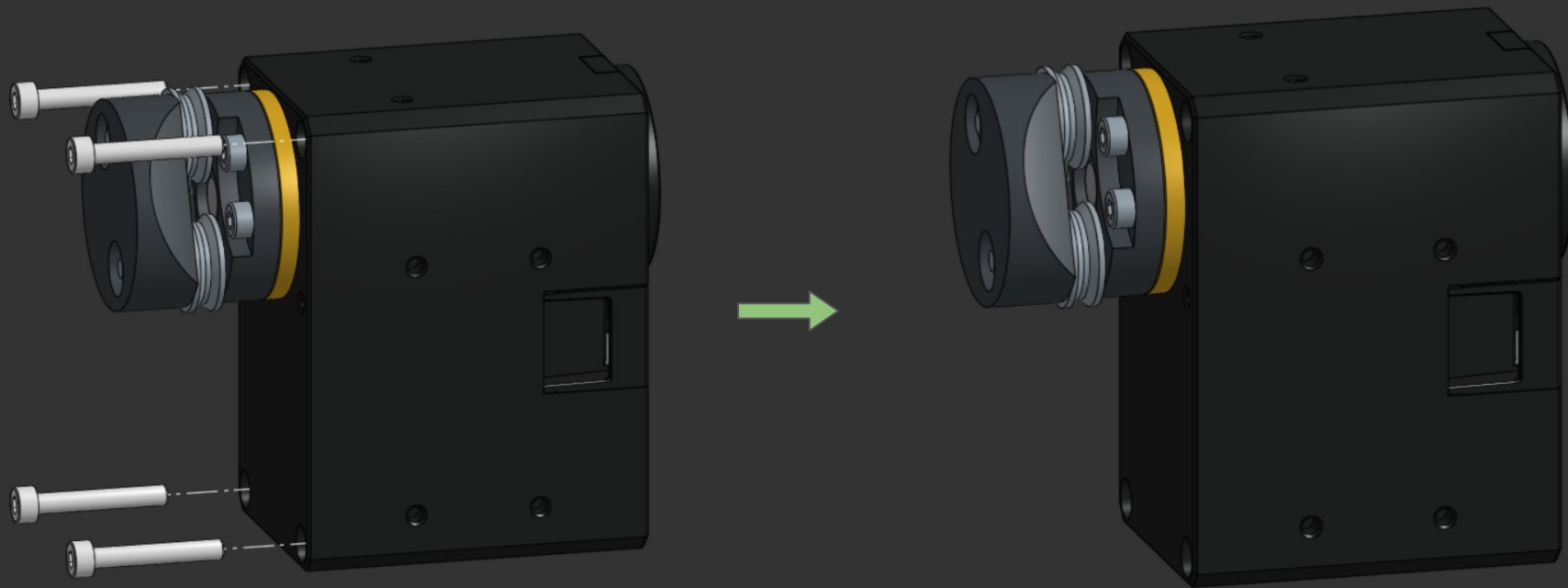
6x M3 Nut

Major Components



Step 1

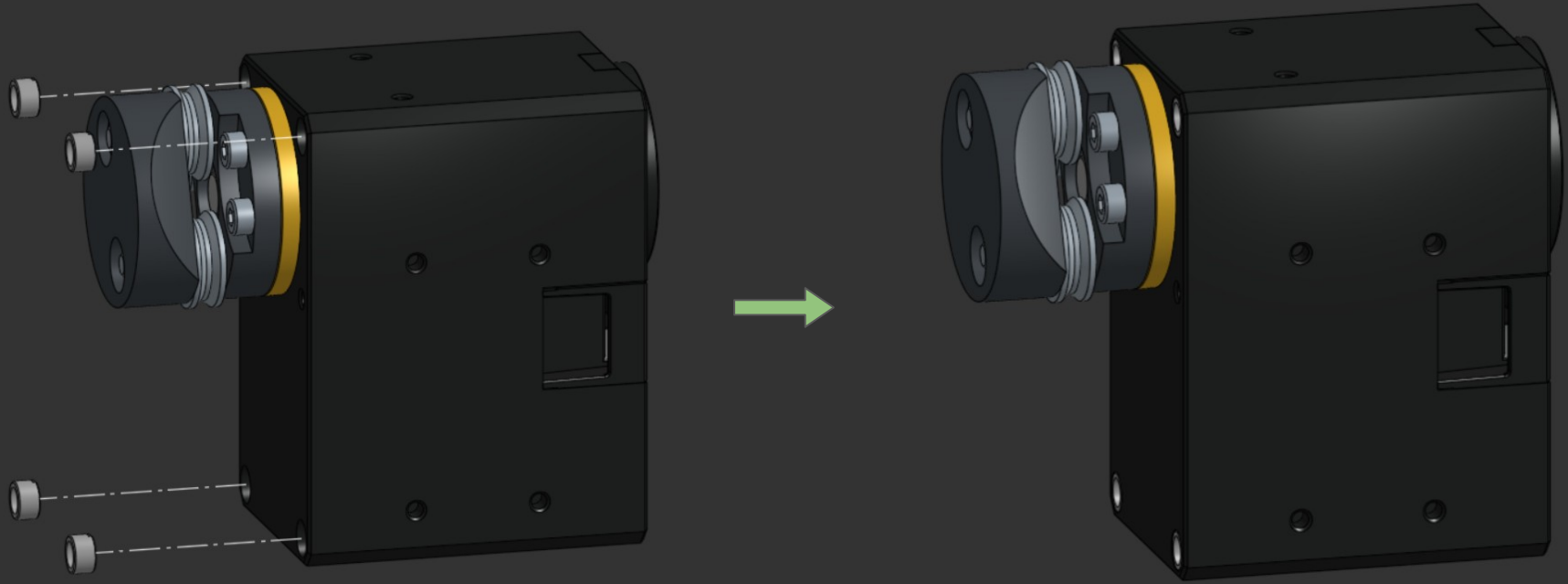
Remove the front four bolts from the actuator



Parts
-Coiler Assembly

Step 2

Insert Robotis Spacers into the front of the actuator



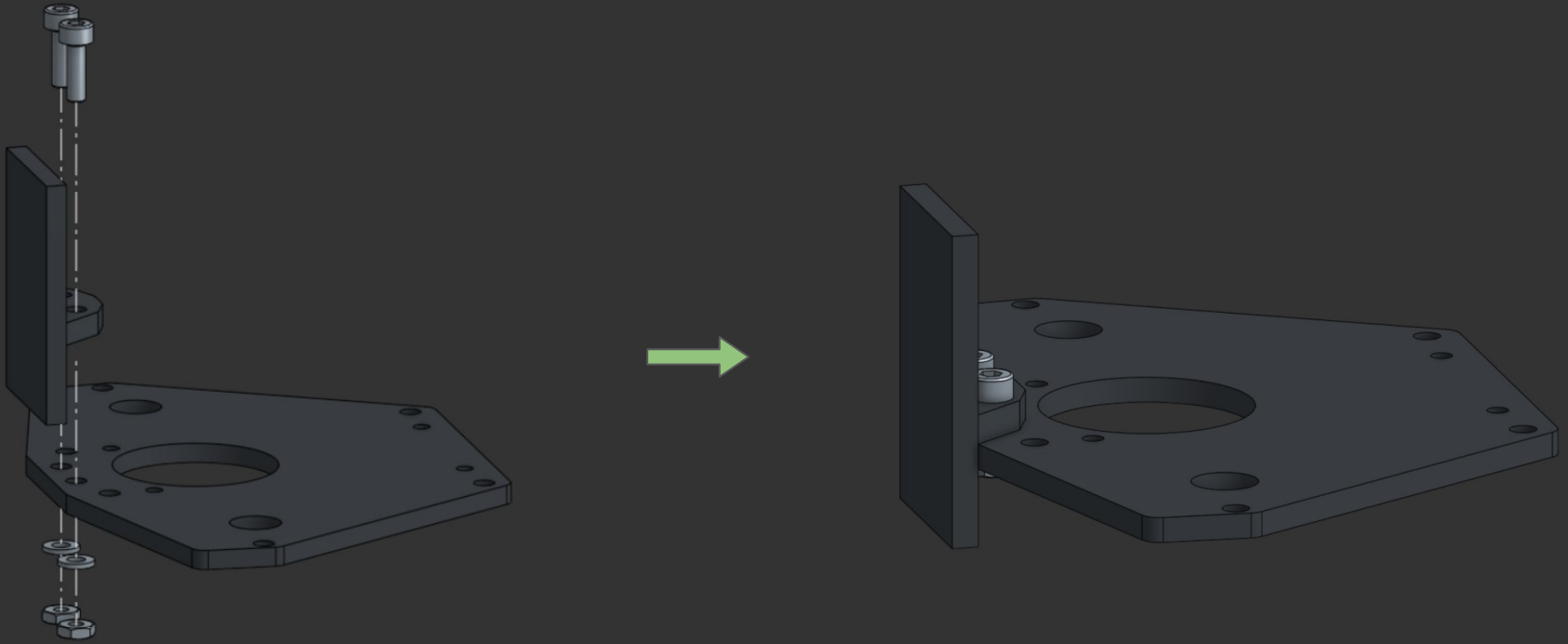
Parts

-Previous step

-4 Robotis Spacers

Step 3

Bolt the Front Finger Plates to the Front Finger Pad with a lock washer underneath the nut



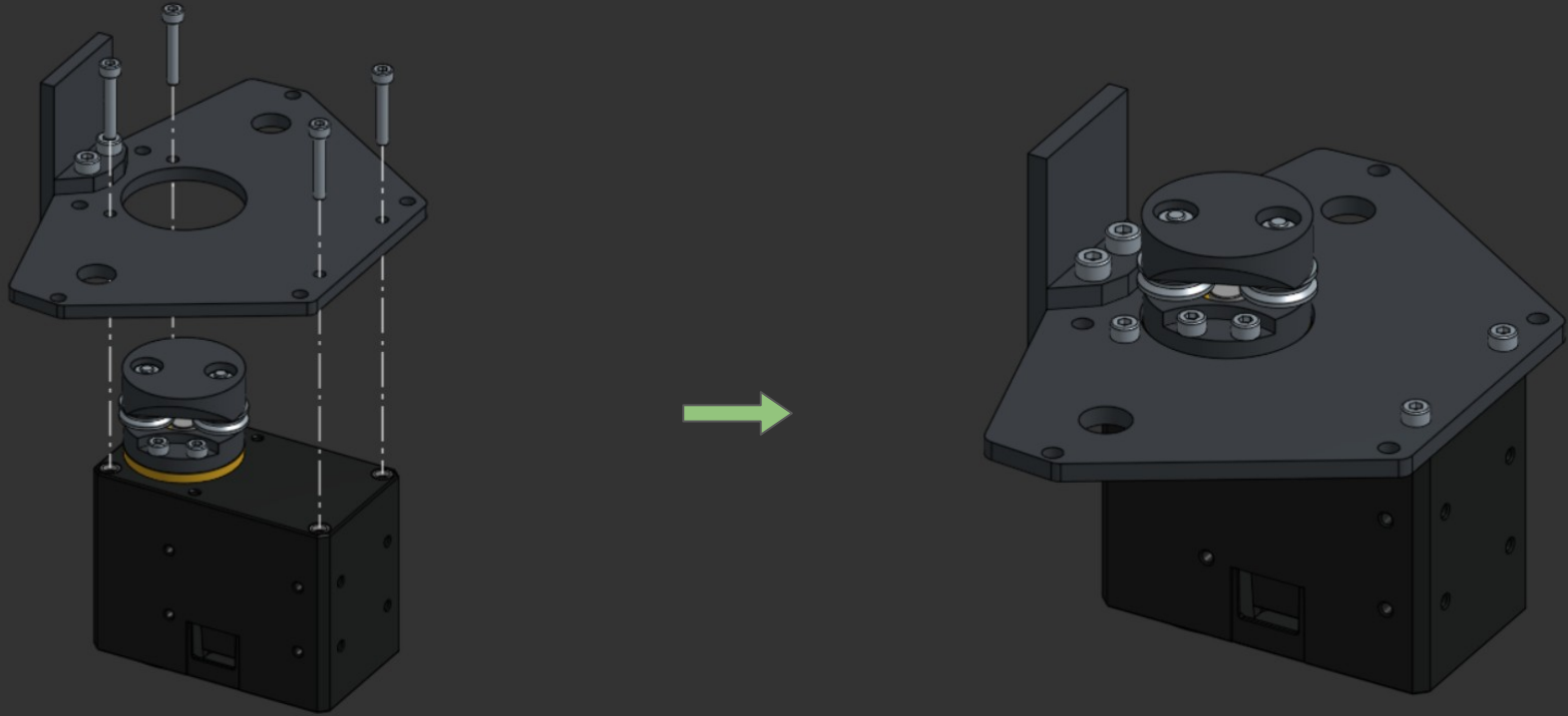
Parts

- Gripper Base Bottom Plate
- Gripper Base Front
- 2x M3x10mm Socket Head Bolt

-2x M3 Lock Washer

Step 4

Bolt the Gripper Base Bottom Plate to the actuator

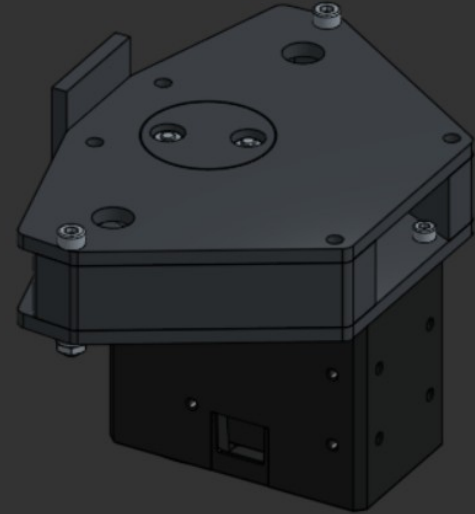
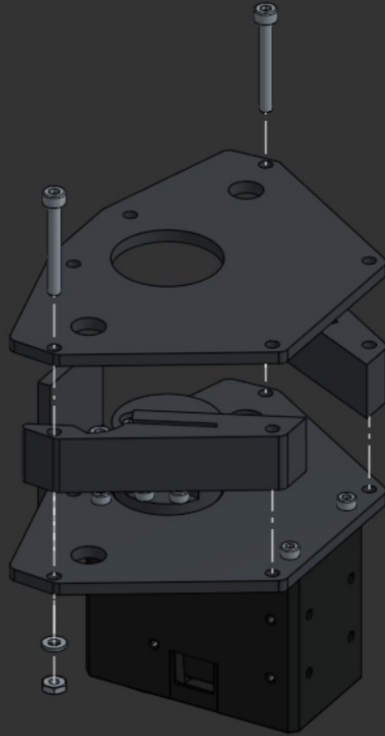


Parts

- Previous step
- Coiler Assembly with spacers
- 4x M2.5x16mm Socket Head Bolt

Step 5

Bolt the Front Finger Plates to the Front Finger Pad with a lock washer underneath the nut

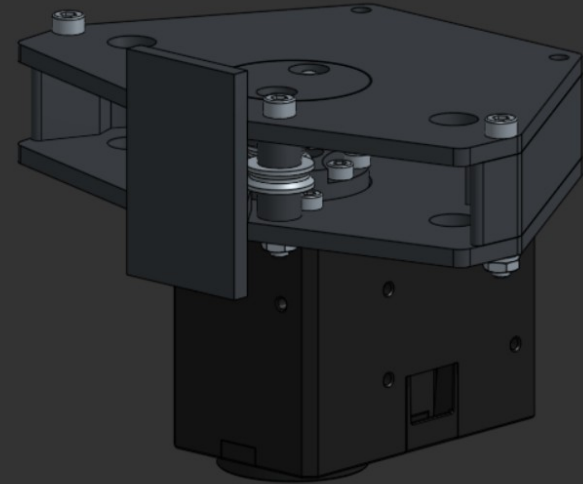
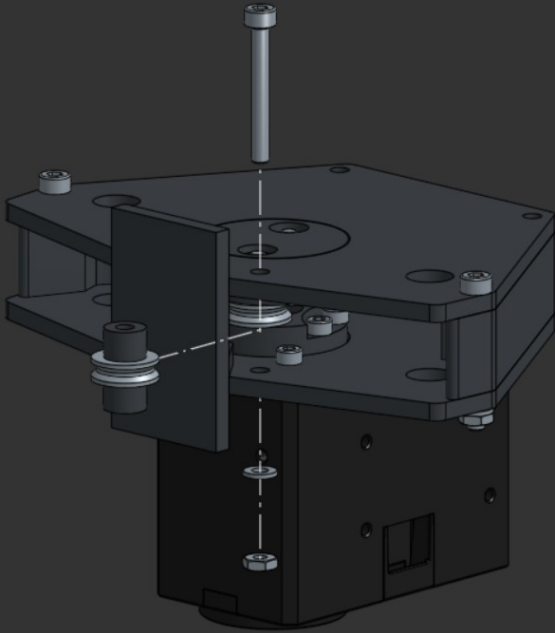


Parts

- Previous step
- Gripper Base Side Wall
- 2x M3x25mm Socket Head Bolt

- 2x Lock Washer
- 2x M3 Nut

Step 6 Bolt the pulley and pulley spacers to the front of the Gripper Base



Parts
-Previous step
-3x12x4mm Pulley
-2x Pulley Spacer

-M3x25mm Socket Head Bolt
-M3 Lock Washer
-M3 Nut

Repeat for the other side

Parts

-Previous step

-3x12x4mm Pulley

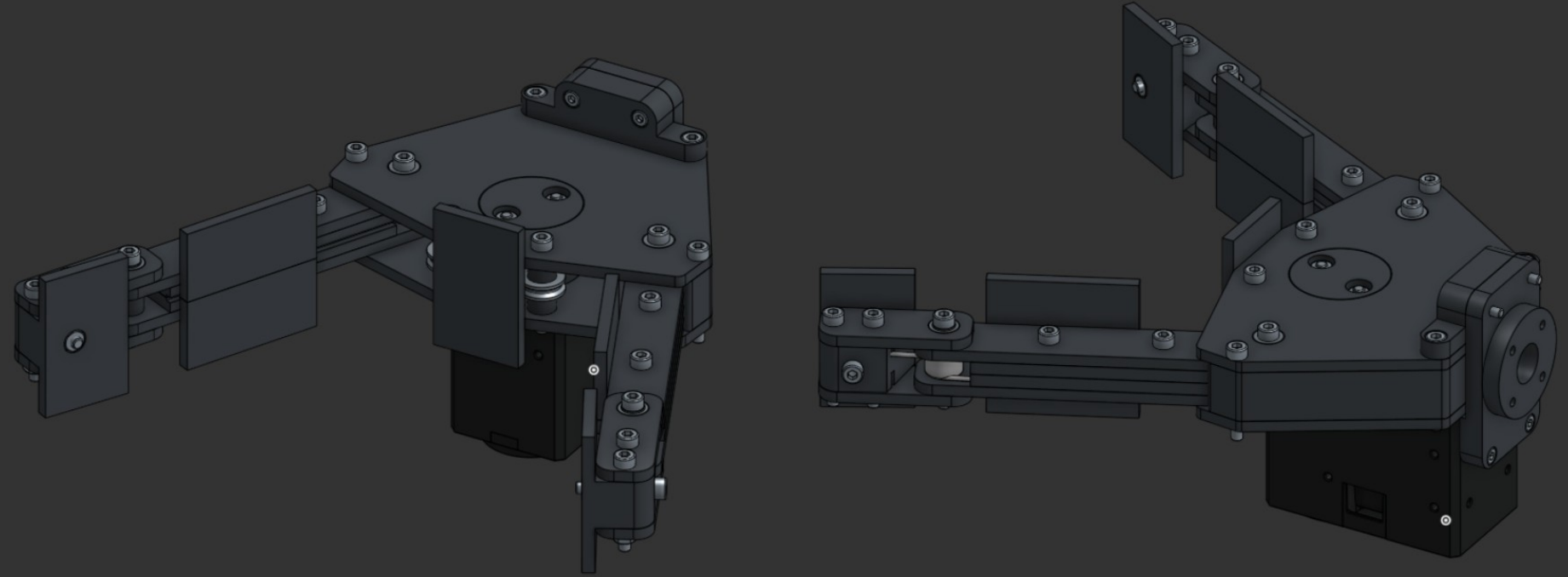
-2x Pulley Spacer

-M3x25mm Socket Head Bolt

-M3 Lock Washer

-M3 Nut

Arm Gripper Assembly Instructions



Parts

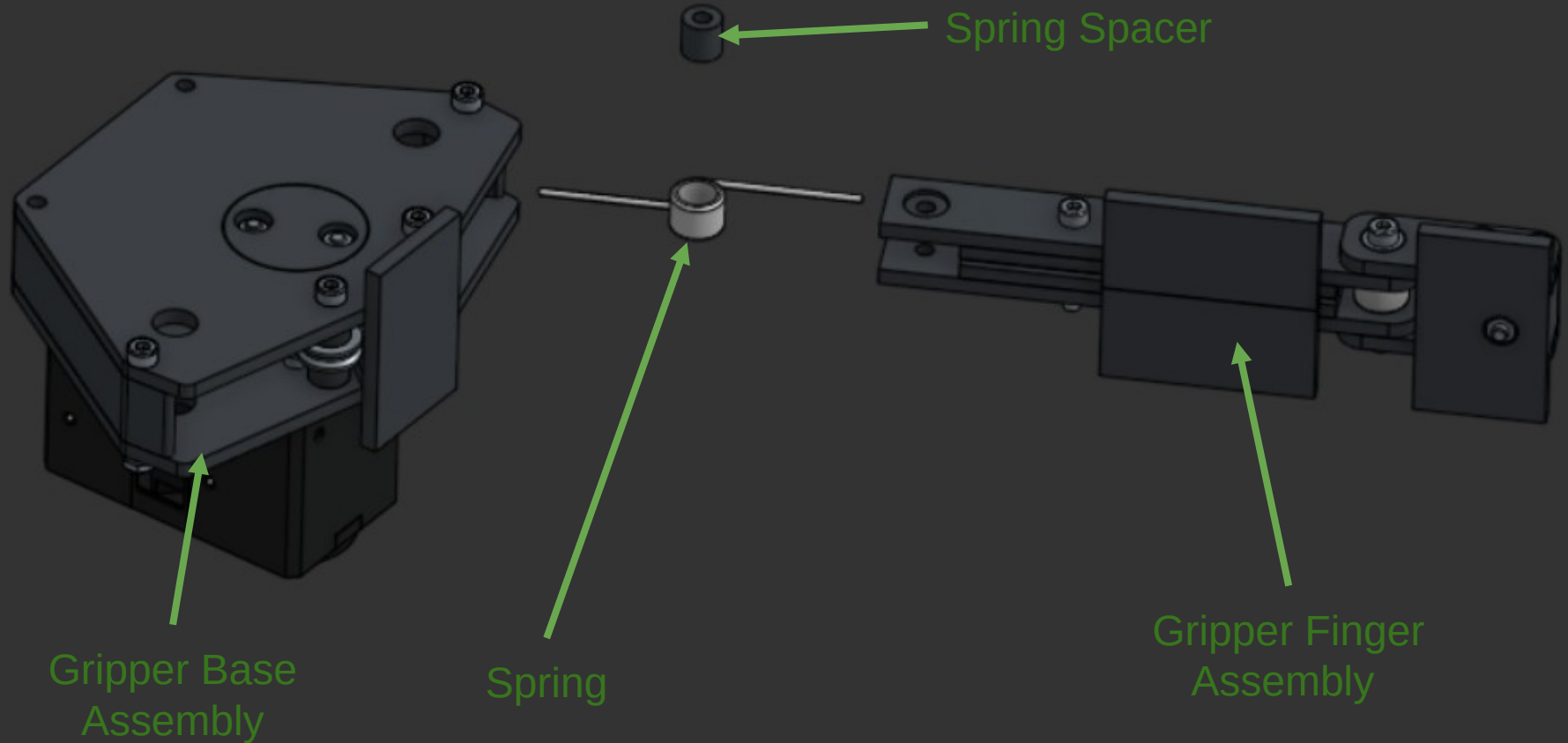
Major Components:

Gripper Base Assembly
2x Gripper Finger Assembly
2x Spring
2x Spring Spacer

Minor Components:

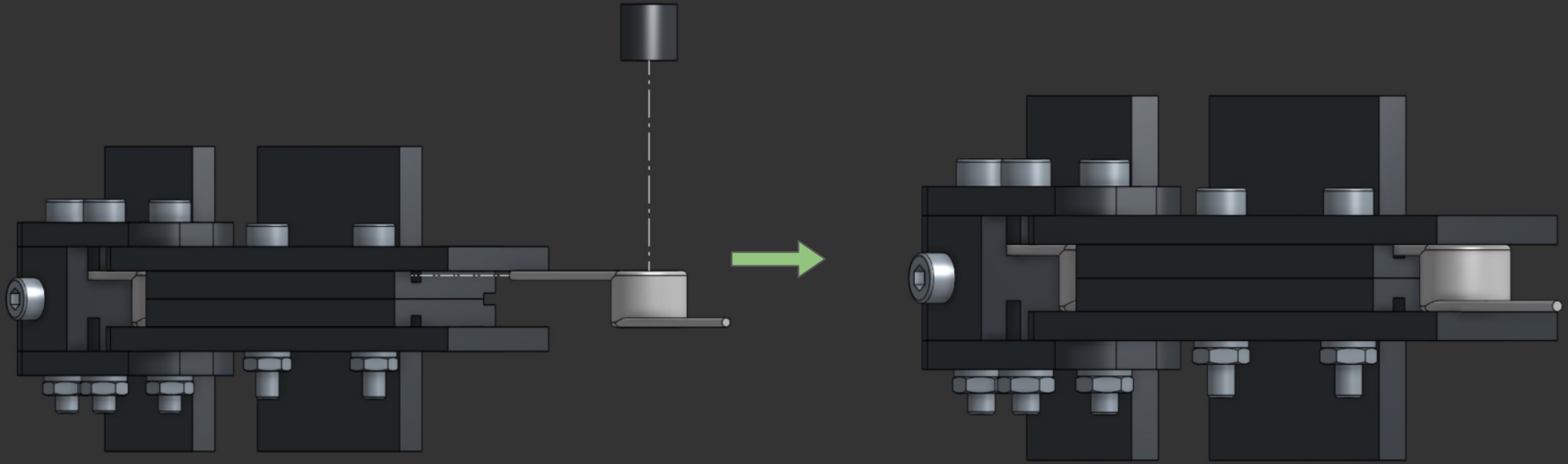
2x M3x25mm Socket Head Bolt
2x M3 Lock Washer
2x M3 Nut

Major Components



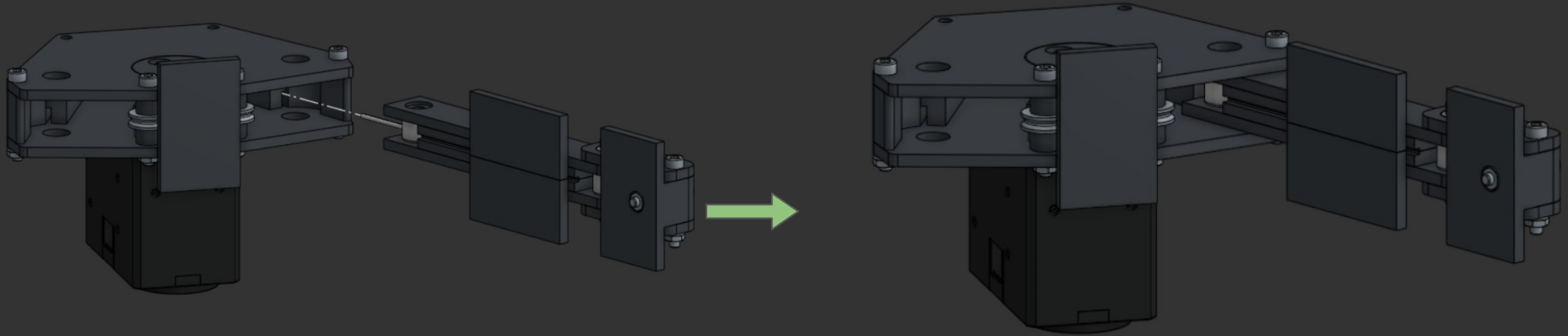
Step 1

Insert a spring spacer into the spring and align the spring with the end of the finger assembly



- Parts
- Gripper Finger Assembly
 - Spring
 - Spring Spacer

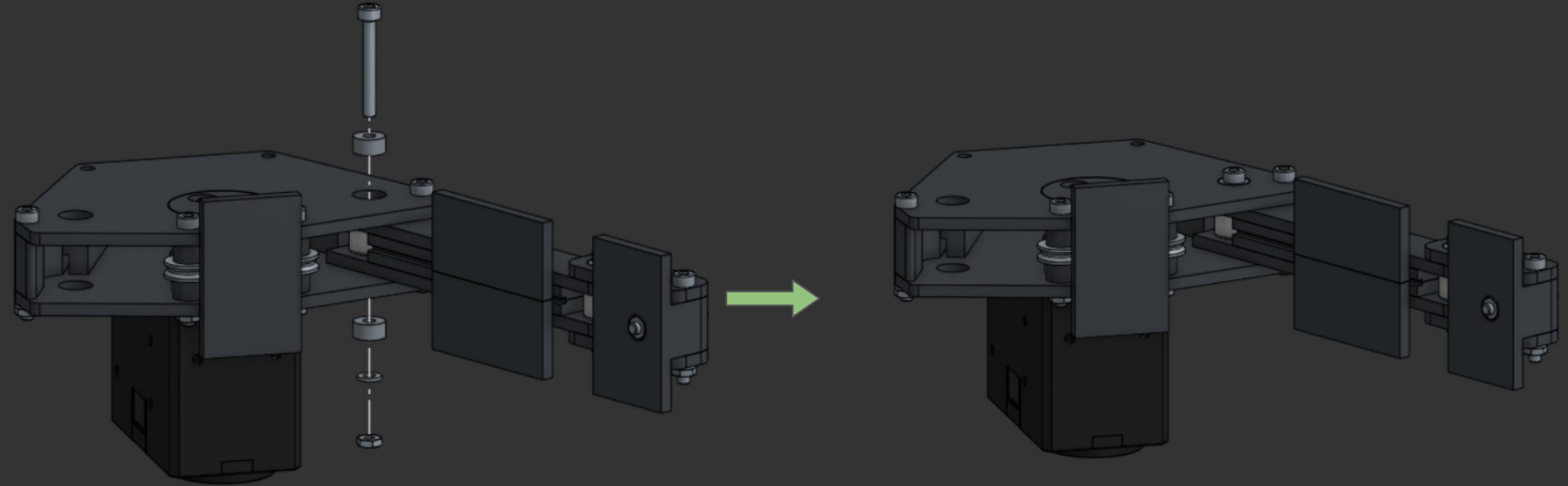
Step 2 Insert the end of the spring into the Gripper Base Side Wall



- Parts
- Previous step
 - Gripper Base Assembly

Step 3

Bolt the finger and two bearings to the gripper base with a lock washer and nut underneath



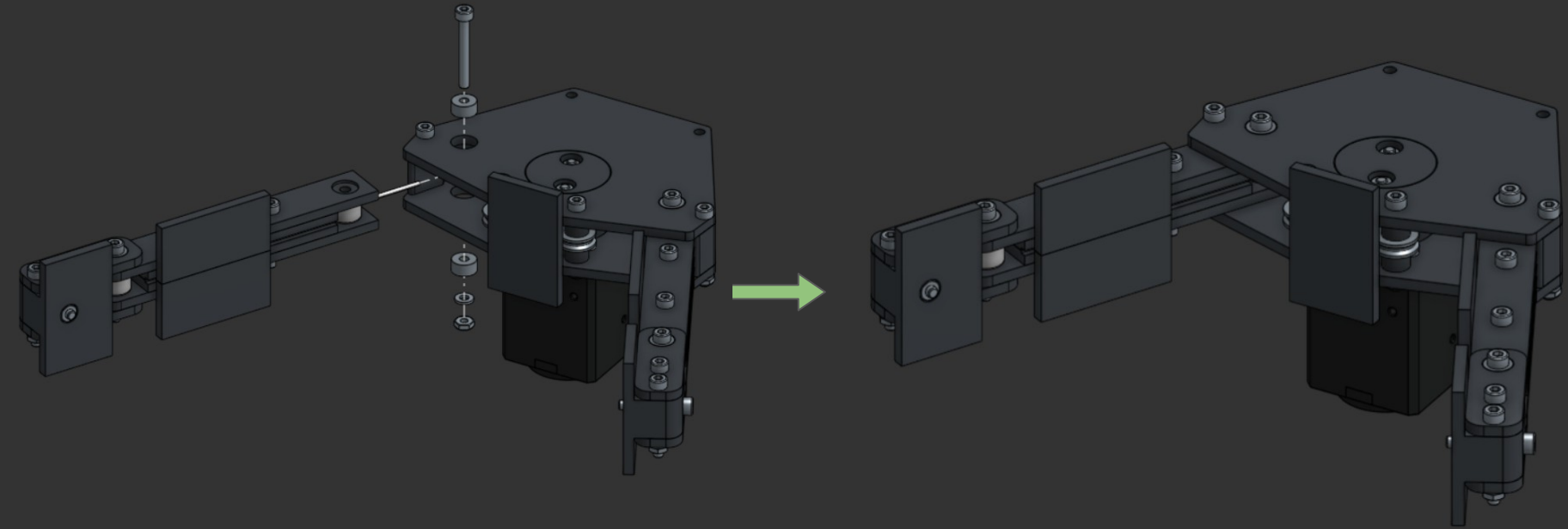
Parts

- Previous step
- M3x25mm Socket Head Bolt
- 2x 3x8x4mm Bearing

- M3 Lock Washer
- M3 Nut

Step 4

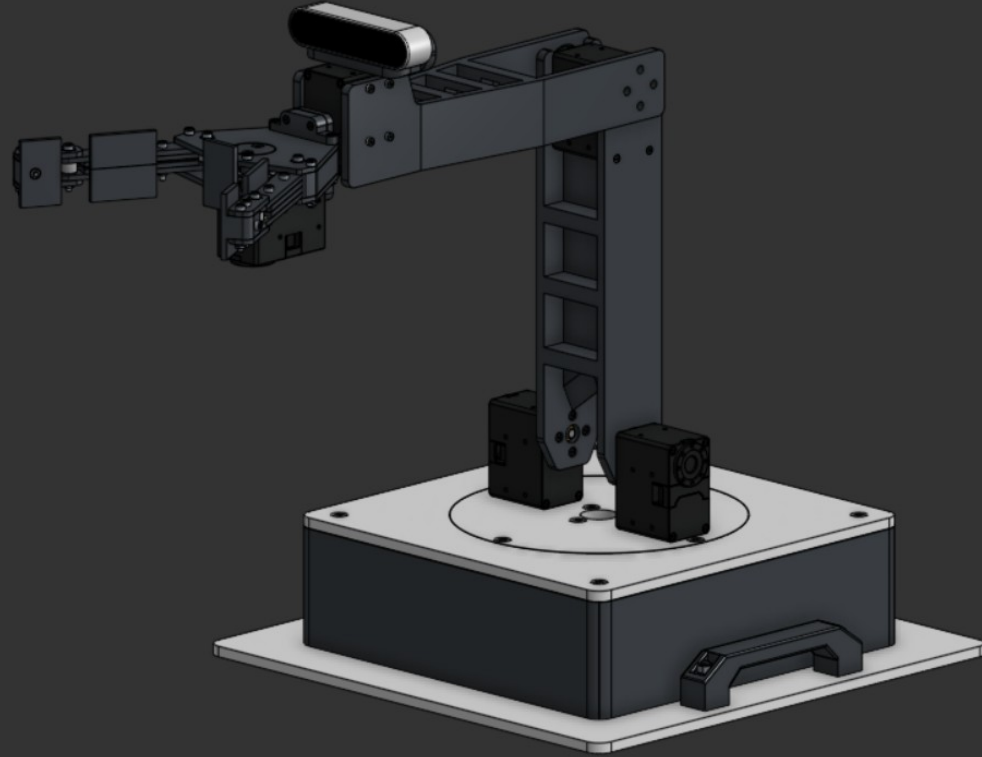
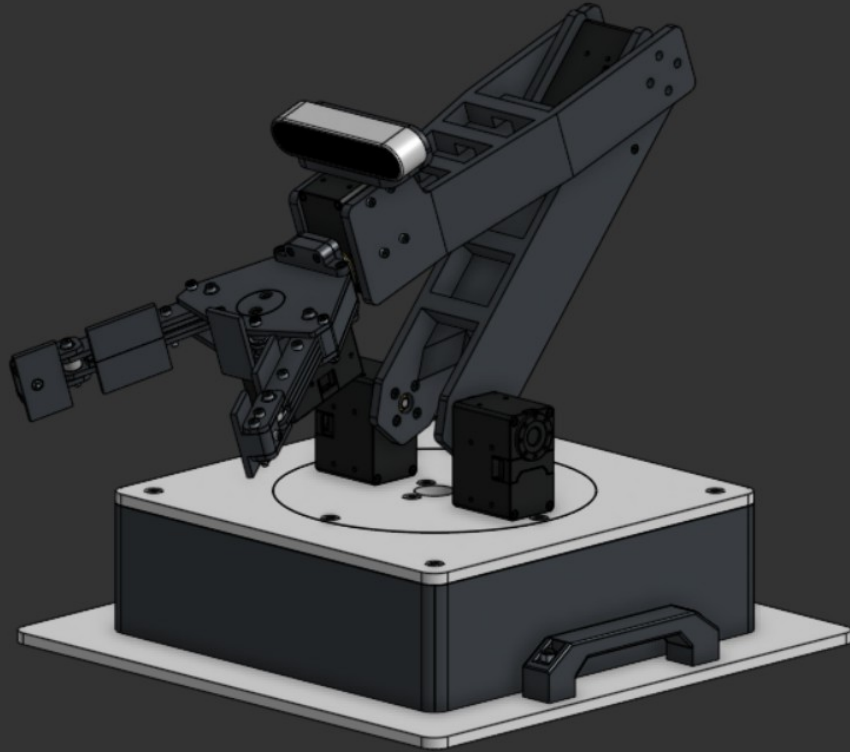
Repeat steps 1-3 for the other finger



Parts
-Previous step
-Gripper Finger Assembly
-M3x25mm Socket Head Bolt

-2x 3x8x4mm Bearing
-M3 Lock Washer
-M3 Nut

Arm Assembly Instructions



Parts

Major Components:

Gripper Assembly

Base Assembly

Shoulder Assembly

Elbow Assembly

Intel Realsense D435i Camera

Camera Mount

Upper Wrist Mount

Center Wrist Mount

Minor Components:

14x M2.5x6mm Socket Head Bolt

10x M2.5x8mm Socket Head Bolt

2x M2.5x14mm Socket Head Bolt

2x M3x25mm Socket Head Bolt

2x M3x10mm Socket Head Bolt

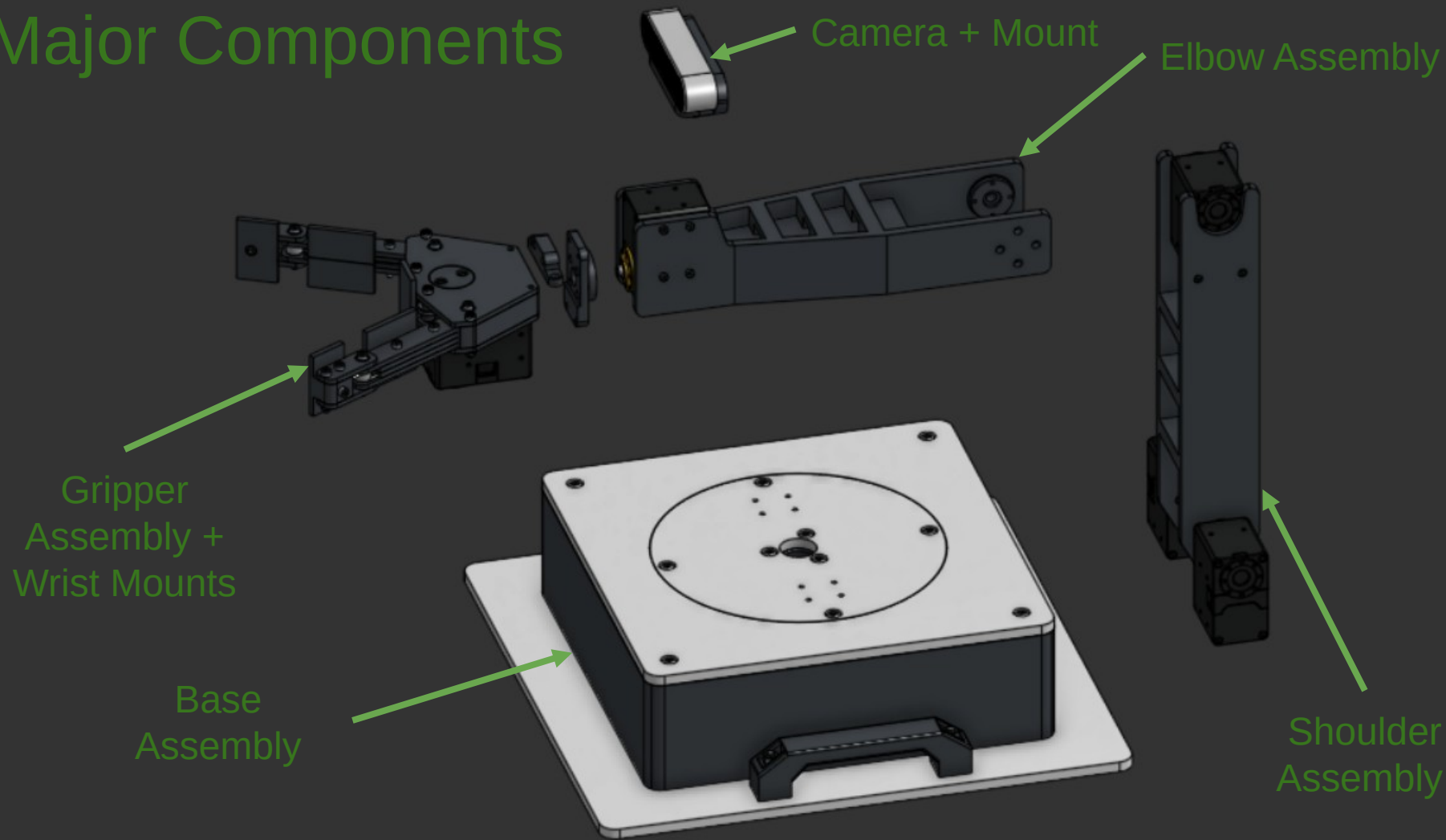
2x M3 Lock Washer

2x M3 Nut

2x M2.5 Lock Washer

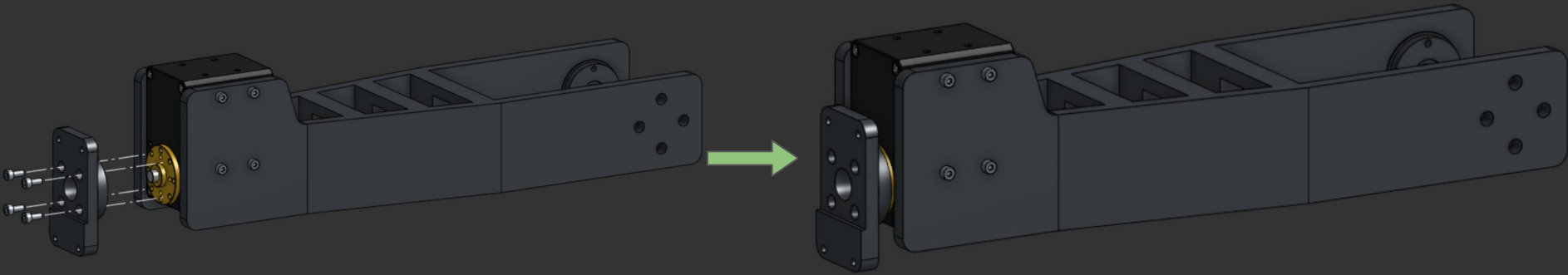
2x M2.5 Nut

Major Components



Step 1

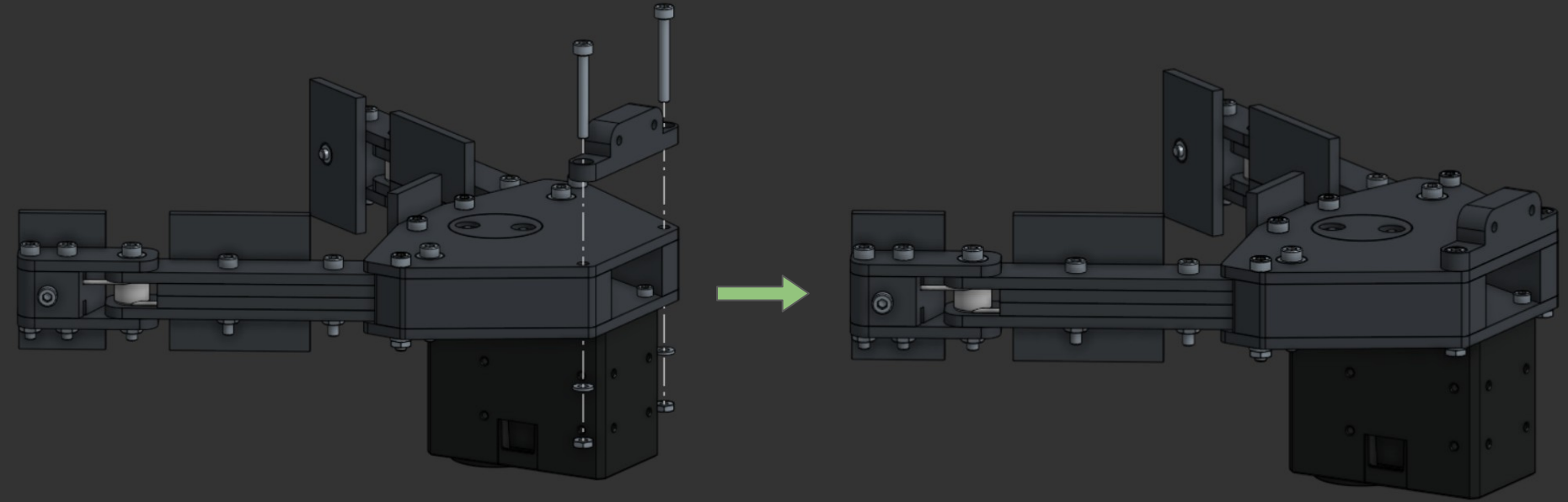
Attach the Center Wrist mount to the Elbow Assembly using 4 bolts



- Parts
- Elbow Assembly
 - Center Wrist Mount
 - 4x M2.5x6mm Socket Head Bolts

Step 2

Bolt the Upper Wrist Mount to the Gripper Assembly



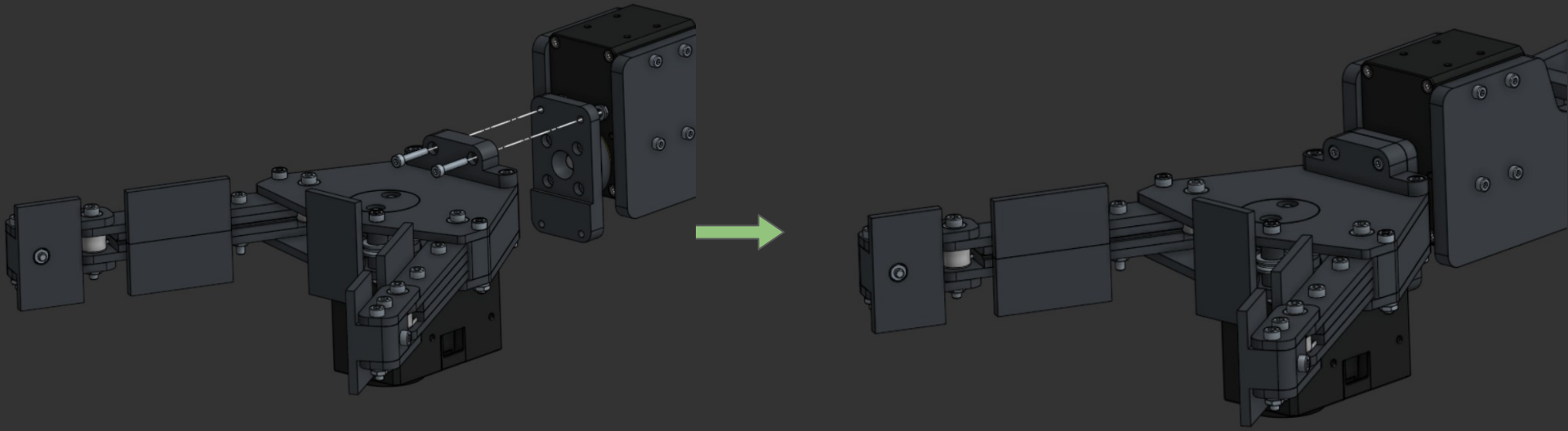
Parts

- Top Wrist Mount
- Gripper Assembly
- 2x M3x25mm Socket Head Bolts

- 2x M3 Lock Washer
- 2x M3 Nut

Step 3

Bolt the Upper Wrist Mount to the Center Wrist Mount



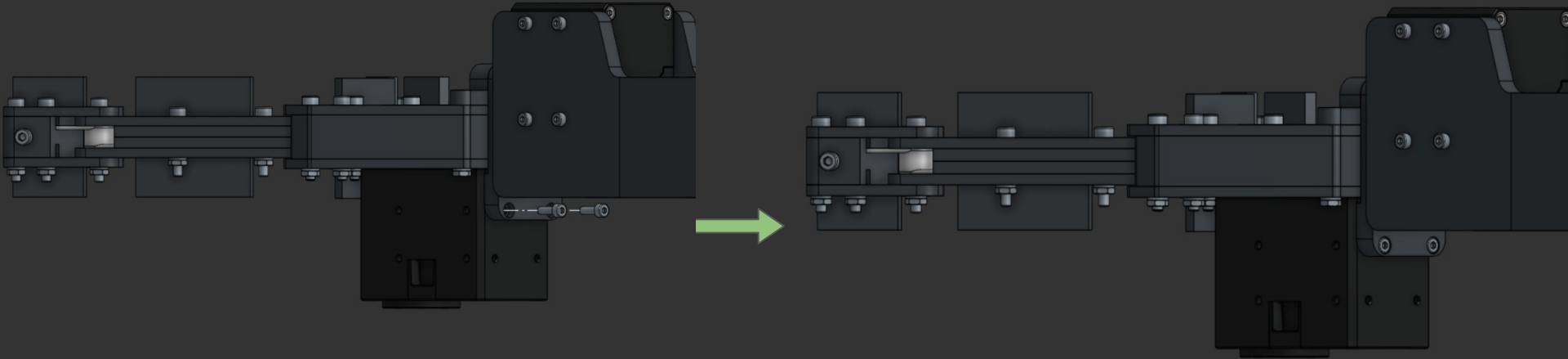
Parts

- Previous step
- 2x M2.5x14mm Socket Head Bolt
- 2x M2.5 Lock Washer

-2x M2.5 Nut

Step 4

Bolt the bottom of the Center Wrist Mount to the gripper actuator



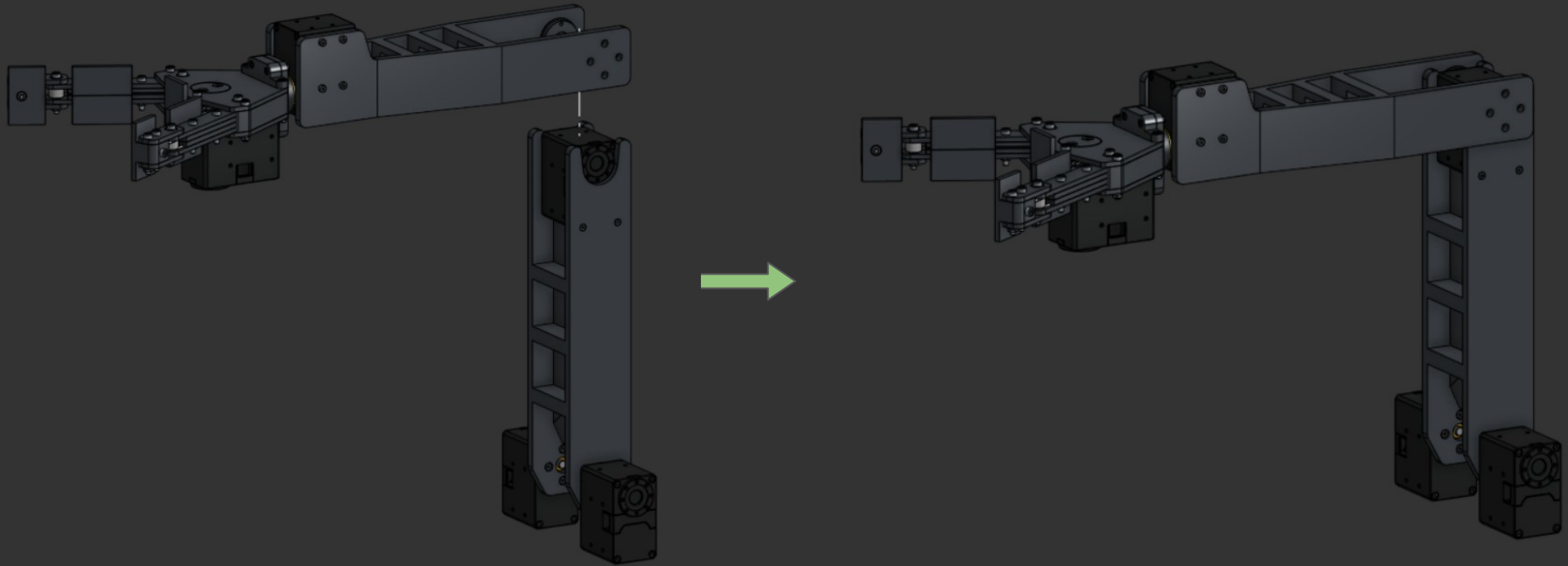
Parts

-Previous step

-2x M2.5x8mm Socket Head Bolt

Step 5

Align the Elbow Assembly with the Shoulder Assembly



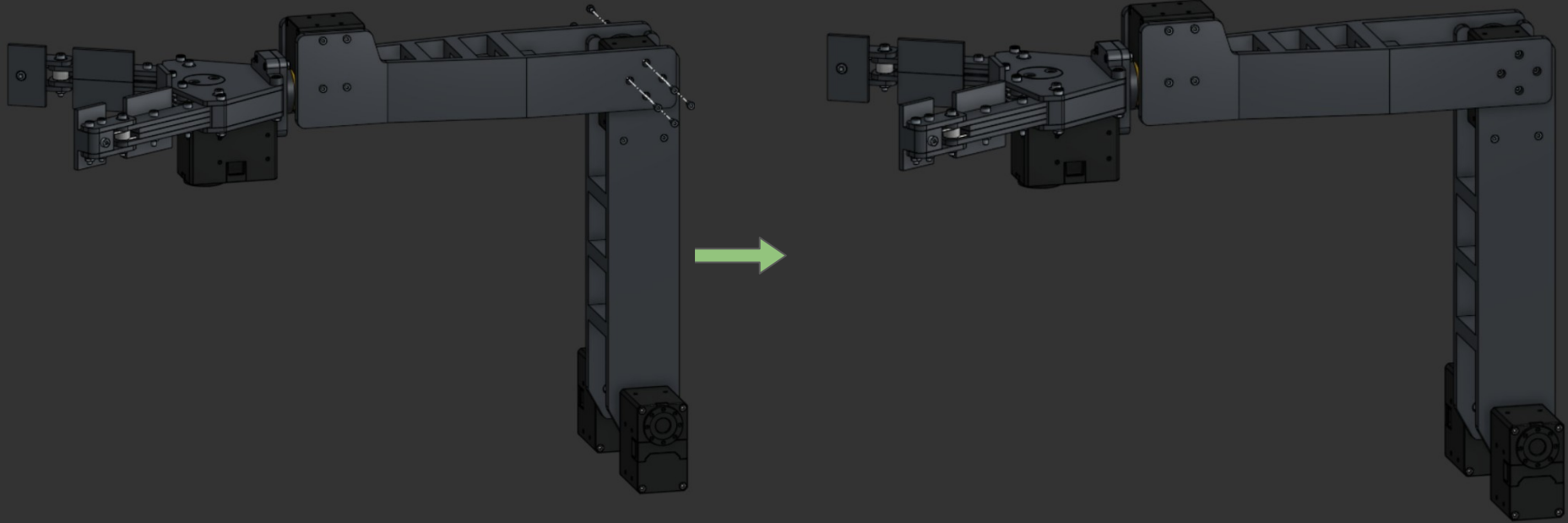
Parts

-Previous step

-Shoulder Assembly

Step 6

Bolt the Elbow Assembly to the Shoulder Assembly



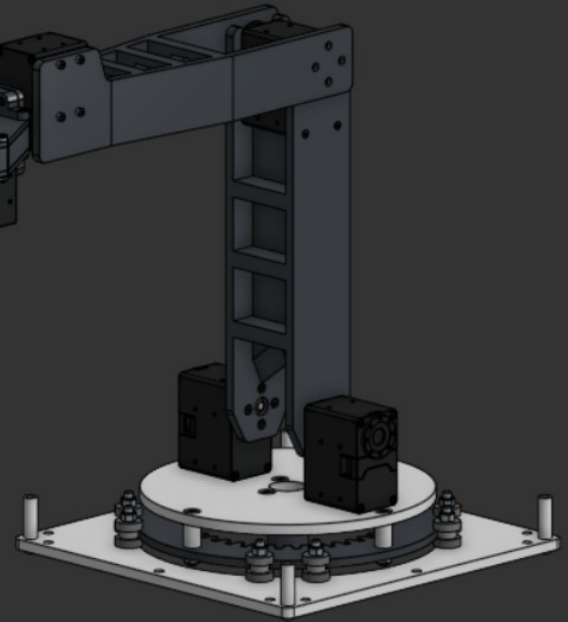
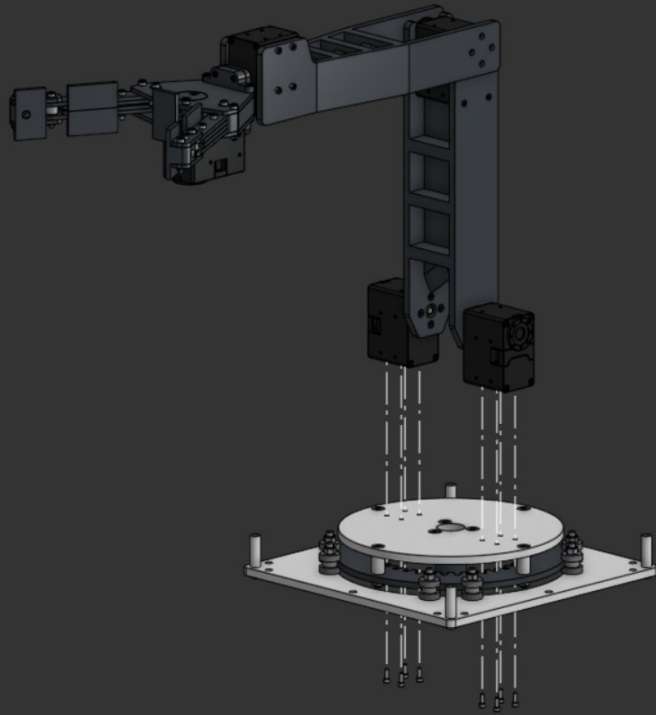
Parts

-Previous step

-8x M2.5x6mm Socket Head Bolt

Step 7

Bolt the Shoulder Assembly to the Arm Base Assembly

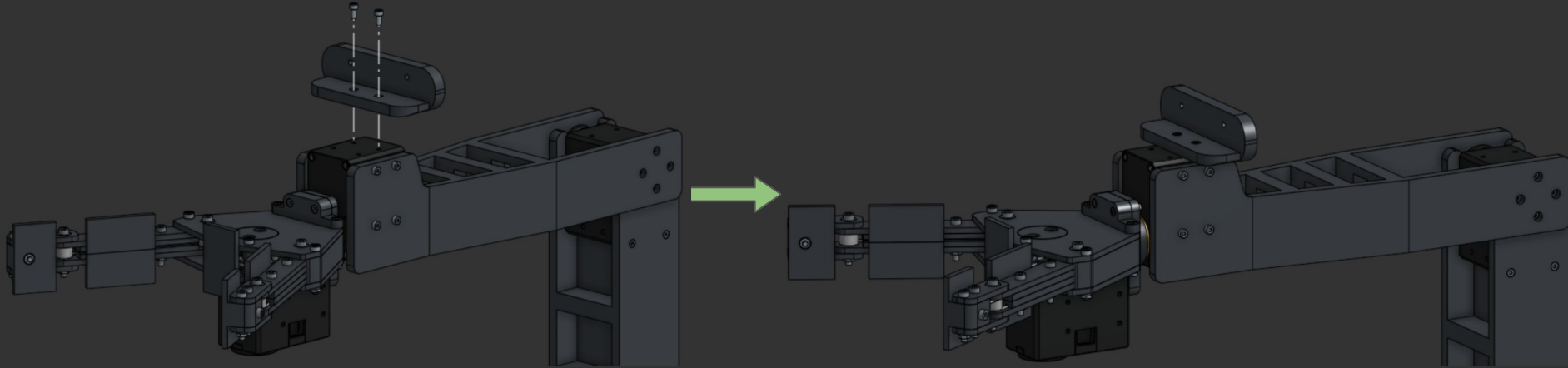


Parts

- Previous step
- Arm Base Assembly
- 8x M2.5x8mm Socket Head Bolt

Step 8

Bolt the Camera Mount to the back two holes of the elbow actuator

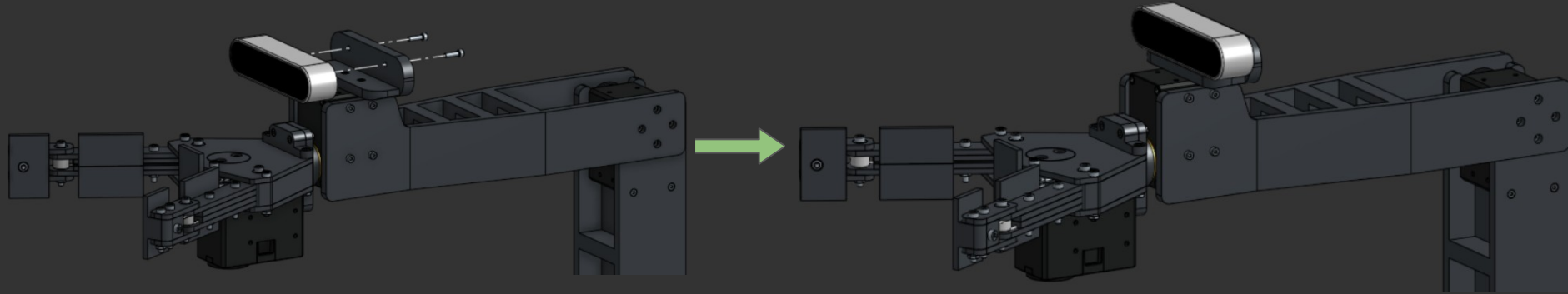


Parts

- Previous step
- Camera Mount
- 2x M2.5x6mm Socket Head Bolt

Step 9

Bolt the camera to the Camera Wrist Mount



Parts

- Previous step
- Intel Realsense D435i Camera
- 2x M3x10mm Socket Head Bolt